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Technologies
4th Year, 1st Semester

Assignment 2

Docker

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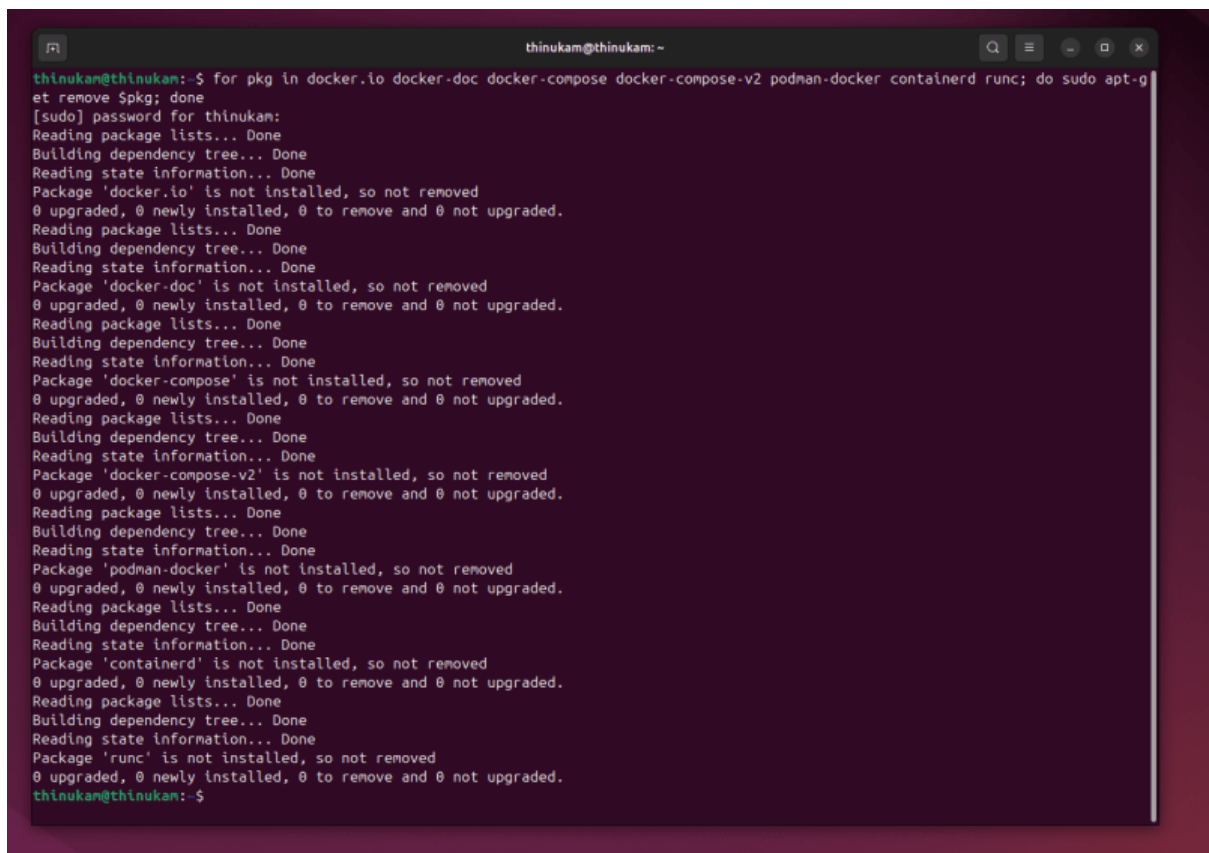
Introduction

This assignment focuses on implementing core network services using Docker containers on Ubuntu 24.04 LTS. The objective is to gain practical experience in containerization by deploying and managing essential infrastructure services such as DNS, DHCP, and Proxy within isolated, lightweight environments. Each service is built using Docker images, configured to meet specific requirements, and then tested to verify proper functionality across the local network. The assignment also emphasizes best practices such as resource limiting, persistent storage through volumes, and centralized deployment using Docker Compose. Finally, the created images are pushed to Docker Hub to demonstrate container portability and reuse across different systems.

1. Initial Setup

1.1.Docker Installation

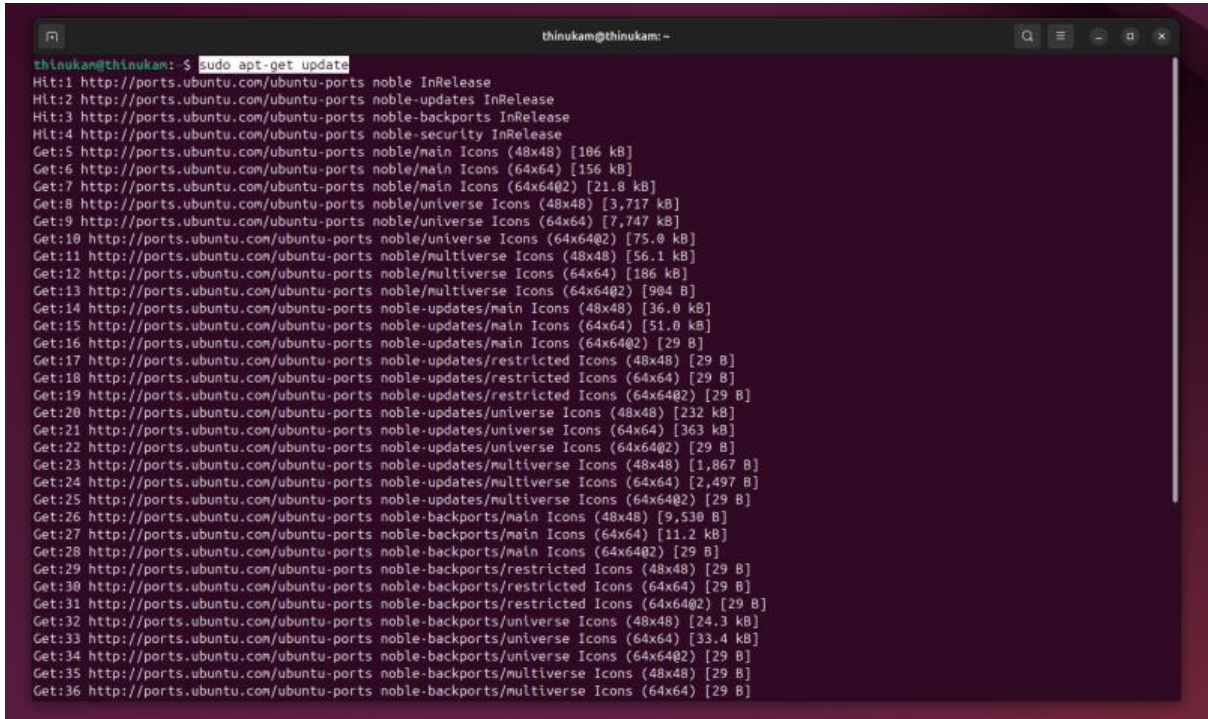
- Linux distributions may offer unofficial Docker packages that may conflict with the official packages provided by Docker.
- It is advisable to uninstall these packages prior to installing the official version of Docker Engine.
- Run the command *for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove \$pkg; done* as given in *Figure 1.1.1*.

A terminal window with a dark background and light green text. The window title is 'thinukam@thinukam: ~'. The command being executed is 'for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove \$pkg; done'. The output shows the process of removing each package, with status messages like 'Package \'docker.io\' is not installed, so not removed' and '0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.' for each package. The prompt returns to 'thinukam@thinukam:~\$' at the end.

```
thinukam@thinukam:~$ for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do sudo apt-get remove $pkg; done
[sudo] password for thinukam:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'docker.io' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'docker-doc' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'docker-compose' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'docker-compose-v2' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'podman-docker' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'containerd' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'runc' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
thinukam@thinukam:~$
```

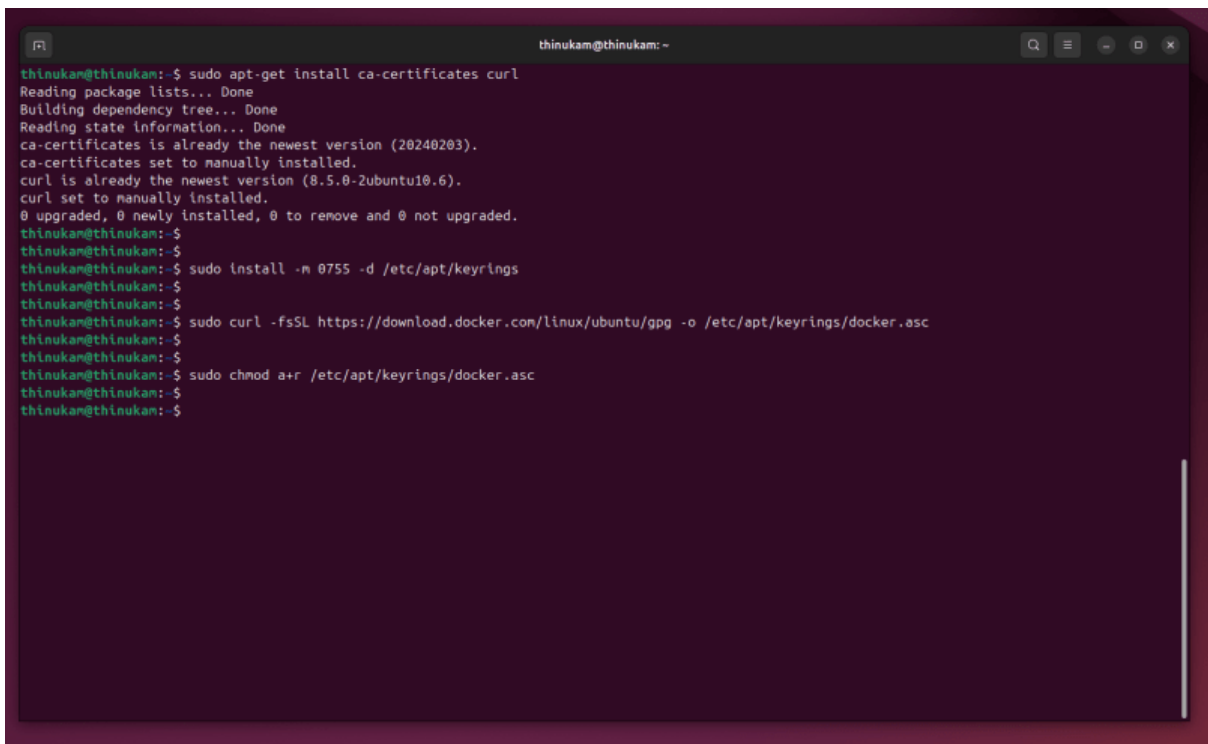
Figure 1.1. 1 - Uninstall conflicting packages

- Prior to installing Docker Engine on a new host machine, it is imperative to configure the Docker *apt* repository.
- To establish Docker's apt repository, run the following commands to include Docker's official GPG key:



```
thinukam@thinukam:~$ sudo apt-get update
Hit:1 http://ports.ubuntu.com/ubuntu-ports noble InRelease
Hit:2 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease
Hit:3 http://ports.ubuntu.com/ubuntu-ports noble-backports InRelease
Hit:4 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/main Icons (48x48) [186 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble/main Icons (64x64) [156 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble/main Icons (64x64@2) [21.8 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble/universe Icons (48x48) [3,717 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports noble/universe Icons (64x64) [7,747 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports noble/universe Icons (64x64@2) [75.0 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports noble/multiverse Icons (48x48) [56.1 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports noble/multiverse Icons (64x64) [186 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports noble/multiverse Icons (64x64@2) [904 B]
Get:14 http://ports.ubuntu.com/ubuntu-ports noble-updates/main Icons (48x48) [36.0 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports noble-updates/main Icons (64x64) [51.0 kB]
Get:16 http://ports.ubuntu.com/ubuntu-ports noble-updates/main Icons (64x64@2) [29 B]
Get:17 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted Icons (48x48) [29 B]
Get:18 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted Icons (64x64) [29 B]
Get:19 http://ports.ubuntu.com/ubuntu-ports noble-updates/restricted Icons (64x64@2) [29 B]
Get:20 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe Icons (48x48) [232 kB]
Get:21 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe Icons (64x64) [363 kB]
Get:22 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe Icons (64x64@2) [29 B]
Get:23 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse Icons (48x48) [1,867 B]
Get:24 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse Icons (64x64) [2,497 B]
Get:25 http://ports.ubuntu.com/ubuntu-ports noble-updates/multiverse Icons (64x64@2) [29 B]
Get:26 http://ports.ubuntu.com/ubuntu-ports noble-backports/main Icons (48x48) [9,530 B]
Get:27 http://ports.ubuntu.com/ubuntu-ports noble-backports/main Icons (64x64) [11.2 kB]
Get:28 http://ports.ubuntu.com/ubuntu-ports noble-backports/main Icons (64x64@2) [29 B]
Get:29 http://ports.ubuntu.com/ubuntu-ports noble-backports/restricted Icons (48x48) [29 B]
Get:30 http://ports.ubuntu.com/ubuntu-ports noble-backports/restricted Icons (64x64) [29 B]
Get:31 http://ports.ubuntu.com/ubuntu-ports noble-backports/restricted Icons (64x64@2) [29 B]
Get:32 http://ports.ubuntu.com/ubuntu-ports noble-backports/universe Icons (48x48) [24.3 kB]
Get:33 http://ports.ubuntu.com/ubuntu-ports noble-backports/universe Icons (64x64) [33.4 kB]
Get:34 http://ports.ubuntu.com/ubuntu-ports noble-backports/universe Icons (64x64@2) [29 B]
Get:35 http://ports.ubuntu.com/ubuntu-ports noble-backports/multiverse Icons (48x48) [29 B]
Get:36 http://ports.ubuntu.com/ubuntu-ports noble-backports/multiverse Icons (64x64) [29 B]
```

Figure I.1. 2 - Update operating system



```
thinukam@thinukam:~$ sudo apt-get install ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
thinukam@thinukam:~$
thinukam@thinukam:~$ sudo install -m 0755 -d /etc/apt/keyrings
thinukam@thinukam:~$
thinukam@thinukam:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
thinukam@thinukam:~$
thinukam@thinukam:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
thinukam@thinukam:~$
```

Figure I.1. 3 - Get official Docker GPG key

- Add the following lines of code to include the repository in the Apt sources and update the system:
 - `echo \`
 - `"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \`
 - `$(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \`
 - `sudo tee /etc/apt/sources.list.d/docker.list > /dev/null`

```
thinukam@thinukam:~$ echo \
> "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
> $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \
> sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
thinukam@thinukam:~$ sudo apt-get update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.8 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable arm64 Packages [31.3 kB]
Hit:3 http://ports.ubuntu.com/ubuntu-ports noble InRelease
Hit:4 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease
Hit:5 http://ports.ubuntu.com/ubuntu-ports noble-backports InRelease
Hit:6 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease
Fetched 80.1 kB in 1s (58.2 kB/s)
Reading package lists... Done
thinukam@thinukam:~$
```

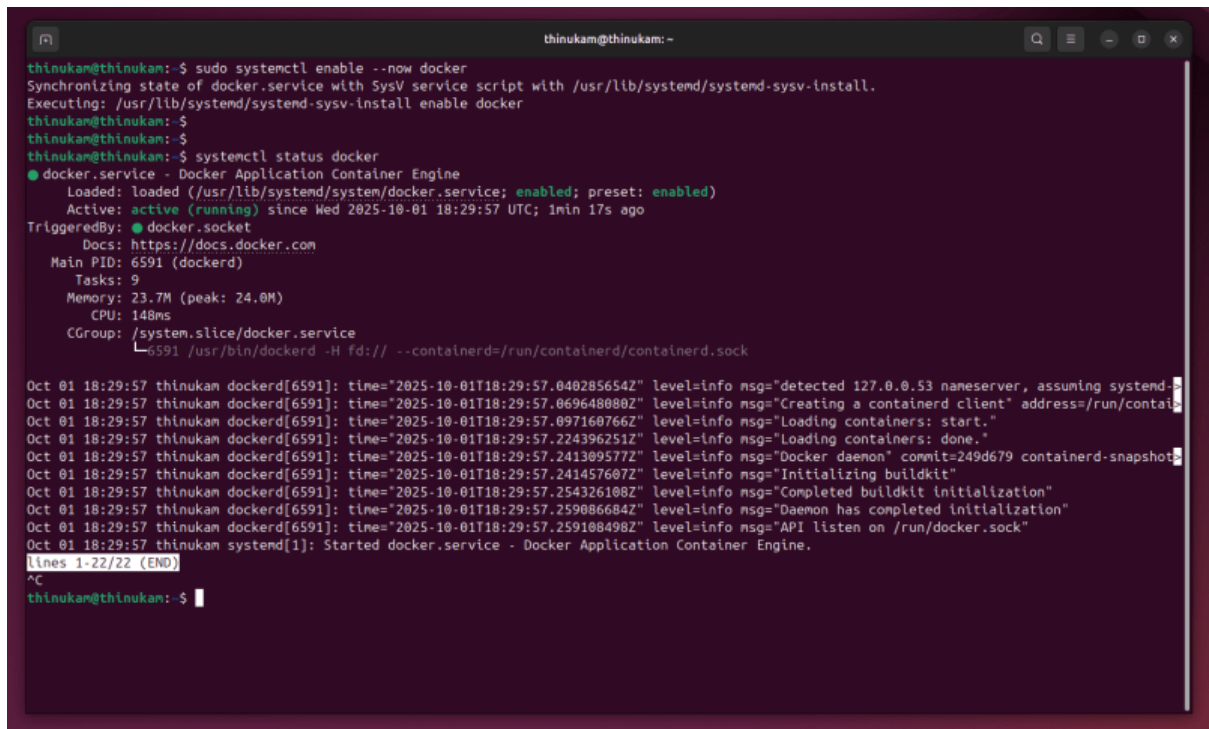
Figure 1.1. 4 - Add the repo to Apt sources

- Upon the successful completion of the preceding steps, proceed with the installation of the latest version of the official Docker Engine packages onto the system.

```
thinukam@thinukam:~$ sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras libslirp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libslirp0 pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 87.5 MB of archives.
After this operation, 397 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 https://download.docker.com/linux/ubuntu noble/stable arm64 containerd.io arm64 1.7.28-0-ubuntu.24.04-noble [23.5 MB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 pigz arm64 2.8-1 [60.7 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 libslirp0 arm64 4.7.0-1ubuntu3 [61.8 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 slirp4netns arm64 1.2.1-1build2 [34.3 kB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable arm64 docker-ce-cli arm64 5:28.4.0-1-ubuntu.24.04-noble [14.9 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable arm64 docker-ce arm64 5:28.4.0-1-ubuntu.24.04-noble [17.0 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable arm64 docker-buildx-plugin arm64 0.29.0-0-ubuntu.24.04-noble [13.9 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable arm64 docker-ce-rootless-extras arm64 5:28.4.0-1-ubuntu.24.04-noble [5,820 kB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable arm64 docker-compose-plugin arm64 2.39.4-0-ubuntu.24.04-noble [12.3 MB]
Fetched 87.5 MB in 9s (9,470 kB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 178063 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_1.7.28-0-ubuntu.24.04-noble_arm64.deb ...
Unpacking containerd.io (1.7.28-0-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a28.4.0-1-ubuntu.24.04-noble_arm64.deb ...
Unpacking docker-ce-cli (5:28.4.0-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a28.4.0-1-ubuntu.24.04-noble_arm64.deb ...
Unpacking docker-ce (5:28.4.0-1-ubuntu.24.04-noble) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1_arm64.deb ...
Unpacking pigz (2.8-1) ...
```

Figure 1.1. 5 - Install Docker packages

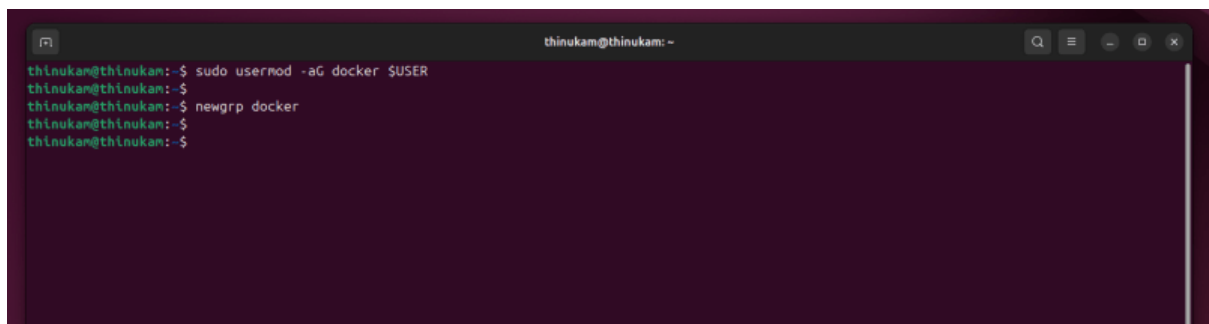
- Initiate the Docker service and configure it to automatically start upon a system reboot.



```
thinukam@thinukam: ~  
thinukam@thinukam:~$ sudo systemctl enable --now docker  
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable docker  
thinukam@thinukam:~$  
thinukam@thinukam:~$ systemctl status docker  
● docker.service - Docker Application Container Engine  
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)  
   Active: active (running) since Wed 2025-10-01 18:29:57 UTC; 1min 17s ago  
TriggeredBy: ● docker.socket  
   Docs: https://docs.docker.com  
  Main PID: 6591 (dockerd)  
    Tasks: 9  
   Memory: 23.7M (peak: 24.0M)  
      CPU: 148ms  
   CGroup: /system.slice/docker.service  
           └─6591 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock  
  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.040285654Z" level=info msg="detected 127.0.0.53 nameserver, assuming systemd-resolved" data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.069648080Z" level=info msg="Creating a containerd client" address=/run/containerd.sock data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.097160766Z" level=info msg="Loading containers: start." data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.224396251Z" level=info msg="Loading containers: done." data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.241309577Z" level=info msg="Docker daemon" commit=249d679 containerd-snapshot= data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.241457607Z" level=info msg="Initializing buildkit" data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.254326108Z" level=info msg="Completed buildkit initialization" data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.259086684Z" level=info msg="Daemon has completed initialization" data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam dockerd[6591]: time="2025-10-01T18:29:57.259108498Z" level=info msg="API listen on /run/docker.sock" data-bbox="117 133 878 457"/>  
Oct 01 18:29:57 thinukam systemd[1]: Started docker.service - Docker Application Container Engine.  
lines 1-22/22 (END)  
^C  
thinukam@thinukam:~$
```

Figure 1.1. 6 - Start and enable Docker

- Add the current user to the Docker group.



```
thinukam@thinukam: ~  
thinukam@thinukam:~$ sudo usermod -aG docker $USER  
thinukam@thinukam:~$  
thinukam@thinukam:~$ newgrp docker  
thinukam@thinukam:~$  
thinukam@thinukam:~$
```

Figure 1.1. 7 - Add user to the Docker group

- Verify the installation is successful by running the *hello-world* image.

```
thinukam@thinukam:~$ docker --version
Docker version 28.4.0, build d8eb465
thinukam@thinukam:~$
thinukam@thinukam:~$
thinukam@thinukam:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
198f93fd5094: Pull complete
Digest: sha256:54e66cc1dd1fcb1c3c58bd8017914dbed8701e2d8c74d9262e26bd9cc1642d31
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (arn64v8)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

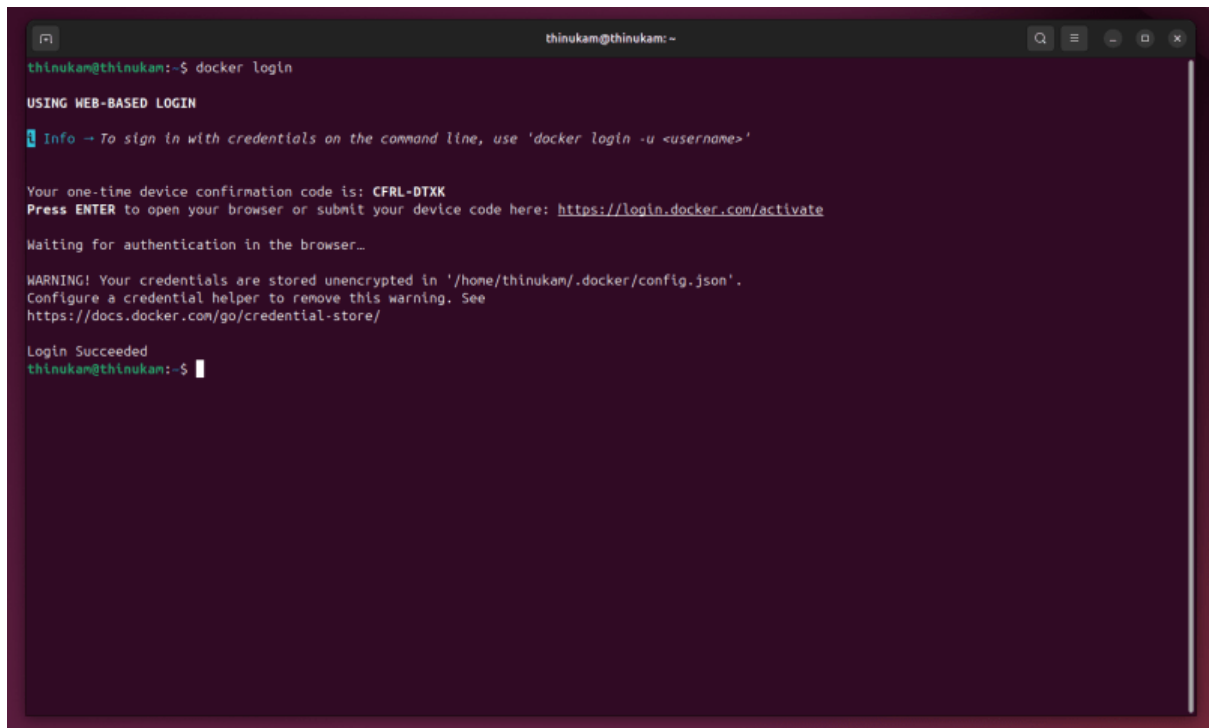
thinukam@thinukam:~$
```

1.2.Docker Hub creation

- Access the docker hub web portal <https://hub.docker.com/> and create a new account.
- Note down the username used when signing up for the docker hub as it will be useful.

Figure 1.2. 1 - Docker Hub sign up

- Log in to Docker Hub via CLI with the command ***docker login***.
- A one-time verification code will be provided along with a Docker login URL. Access the URL and enter the verification code to successfully log in to Docker Hub.

A terminal window with a dark background and light text. The window title is 'thinukam@thinukam: ~'. The user has entered the command 'docker login'. The output shows 'USING WEB-BASED LOGIN' and an info message about signing in with credentials. It then displays a one-time device confirmation code 'CFRL-DTXK' and a URL 'https://login.docker.com/activate'. The user is prompted to press ENTER to open the browser. The terminal shows 'Waiting for authentication in the browser...' and a warning about unencrypted credentials. Finally, it says 'Login Succeeded' and returns to the prompt.

```
thinukam@thinukam:~$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: CFRL-DTXK
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

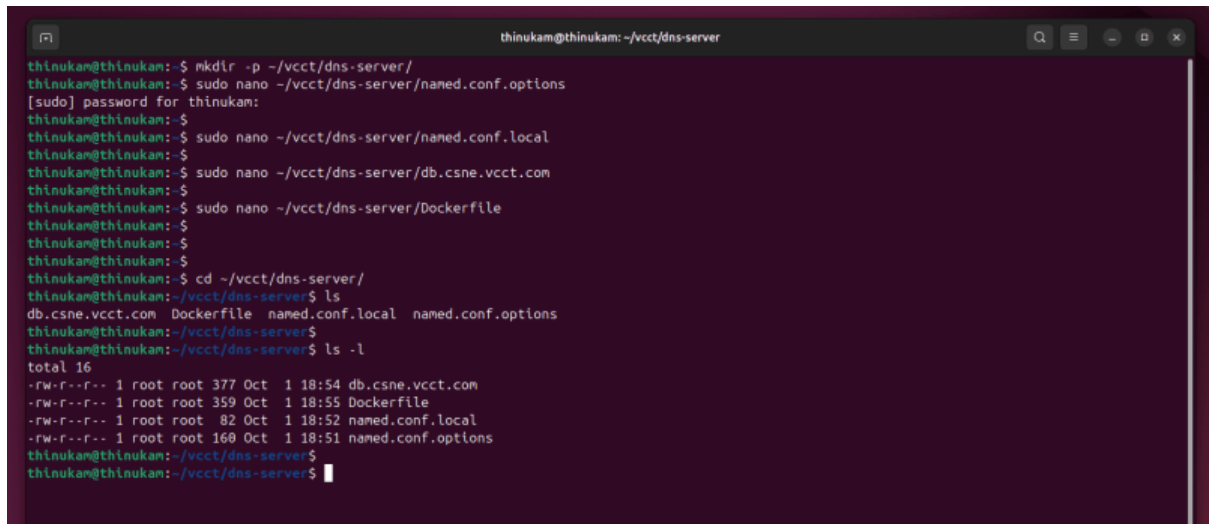
WARNING! Your credentials are stored unencrypted in '/home/thinukam/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
thinukam@thinukam:~$
```

Figure 1.2. 2 – Docker hub login via CLI

2. DNS Server

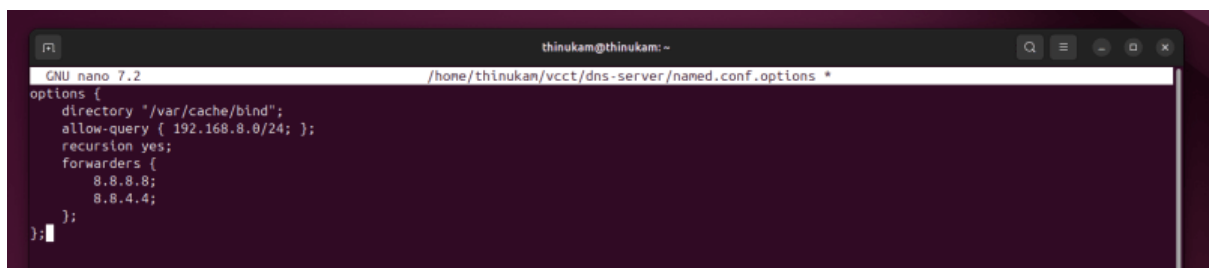
- For the DNS server configuration, several files were created to define the operation of BIND9 within the container.
- The following directory hierarchy is maintained to store the required DNS files for the purpose of creating the Docker image.



```
thinukam@thinukam: ~/vcct/dns-server
thinukam@thinukam:~$ mkdir -p ~/vcct/dns-server/
thinukam@thinukam:~$ sudo nano ~/vcct/dns-server/named.conf.options
[sudo] password for thinukam:
thinukam@thinukam:~$ sudo nano ~/vcct/dns-server/named.conf.local
thinukam@thinukam:~$ sudo nano ~/vcct/dns-server/db.csne.vcct.com
thinukam@thinukam:~$ sudo nano ~/vcct/dns-server/Dockerfile
thinukam@thinukam:~$
thinukam@thinukam:~$ cd ~/vcct/dns-server/
thinukam@thinukam:~/vcct/dns-server$ ls
db.csne.vcct.com  Dockerfile  named.conf.local  named.conf.options
thinukam@thinukam:~/vcct/dns-server$
thinukam@thinukam:~/vcct/dns-server$ ls -l
total 16
-rw-r--r-- 1 root root 377 Oct 1 18:54 db.csne.vcct.com
-rw-r--r-- 1 root root 359 Oct 1 18:55 Dockerfile
-rw-r--r-- 1 root root 82 Oct 1 18:52 named.conf.local
-rw-r--r-- 1 root root 160 Oct 1 18:51 named.conf.options
thinukam@thinukam:~/vcct/dns-server$
thinukam@thinukam:~/vcct/dns-server$
```

Figure 2. 1 - Directory hierarchy for DNS

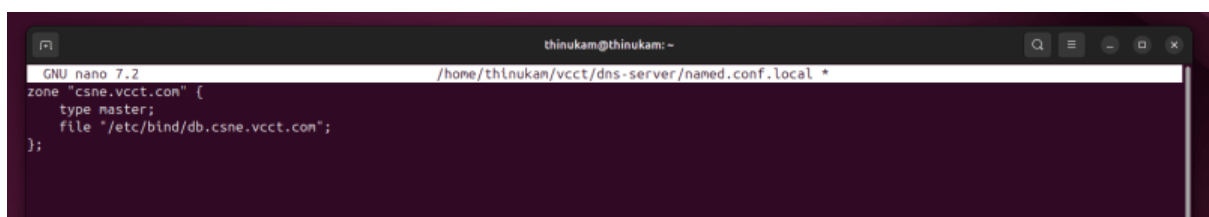
- The following contents were added to the ***named.conf.options*** file to specify global options such as recursion, forwarders, and access control.



```
GNU nano 7.2 /home/thinukam/vcct/dns-server/named.conf.options *
options {
    directory "/var/cache/bind";
    allow-query { 192.168.8.0/24; };
    recursion yes;
    forwarders {
        8.8.8.8;
        8.8.4.4;
    };
};
```

Figure 2. 2 - ***named.conf.options*** configuration file

- The following contents were added to the ***named.conf.local*** file to declare the custom zone for ***csne.vcct.com*** and links it to the appropriate zone file.



```
GNU nano 7.2 /home/thinukam/vcct/dns-server/named.conf.local *
zone "csne.vcct.com" {
    type master;
    file "/etc/bind/db.csne.vcct.com";
};
```

Figure 2. 3 - ***named.conf.local*** configuration file

- The following contents were added to the **db.csne.vcct.com** file to specify the actual DNS source records, including SOA, NS, and A records.

```

GNU nano 7.2 /home/thinukam/vcct/dns-server/db.csne.vcct.com *
$TTL 604800
@      IN      SOA     localhost. root.localhost. (
                        1          ; Serial
                        604800     ; Refresh
                        86400      ; Retry
                        2419200    ; Expire
                        604800 )   ; Negative Cache TTL
;
@      IN      NS     localhost.
@      IN      A      192.168.8.132

```

Figure 2. 4 - db.csne.vcct.com configuration file

- The **Dockerfile** was created to automate the setup of BIND9 within the container by installing the required BIND9 packages, and copying the custom configuration files to **/etc/bind/** directory in the container.

```

GNU nano 7.2 /home/thinukam/vcct/dns-server/Dockerfile *
FROM ubuntu:latest

RUN apt-get update && apt-get install -y bind9 bind9utils bind9-doc dnstools && \
    rm -rf /var/lib/apt/lists/*

COPY named.conf.options /etc/bind/named.conf.options
COPY named.conf.local /etc/bind/named.conf.local
COPY db.csne.vcct.com /etc/bind/db.csne.vcct.com

EXPOSE 53/tcp 53/udp

CMD ["named", "-g", "-c", "/etc/bind/named.conf"]

```

Figure 2. 5 - Dockerfile to automate DNS setup

- Change the current working directory to the directory where the DNS Dockerfile is maintained (**~/vcct/dns-server/** in this instance).
- Build the docker image for the DNS server and verify the successful creation of the image.

```

thinukam@thinukam: ~/vcct/dns-server
thinukam@thinukam:~/vcct/dns-server$ docker build -t it22310132/dns-server:1.0 .
[+] Building 31.4s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 398B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [auth] library/ubuntu:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/ubuntu:latest@sha256:353675e2a41babd526e2b837d7ec780c2a05bca0164f7ea5dbbd433d21d166fc
=> => resolve docker.io/library/ubuntu:latest@sha256:353675e2a41babd526e2b837d7ec780c2a05bca0164f7ea5dbbd433d21d166fc
=> => sha256:353675e2a41babd526e2b837d7ec780c2a05bca0164f7ea5dbbd433d21d166fc 6.69kB / 6.69kB
=> => sha256:d592e55295194d2485a9bd2ce18de6ebf9e6db2fa63d2e264d602e3281bf52b6 424B / 424B
=> => sha256:f415f3f99819fe04db19db0944ba85d0c19160a142ae8ce69006236b446cd18 2.31kB / 2.31kB
=> => sha256:59a5d47f84c39a2d62d1b5089e60ab67303111f17e1df01dbbcc598246282797 28.86MB / 28.86MB
=> => extracting sha256:59a5d47f84c39a2d62d1b5089e60ab67303111f17e1df01dbbcc598246282797
=> [internal] load build context
=> => transferring context: 758B
=> [2/5] RUN apt-get update && apt-get install -y bind9 bind9utils bind9-doc dnstools && rm -rf /var/lib/apt/lists/*
=> [3/5] COPY named.conf.options /etc/bind/named.conf.options
=> [4/5] COPY named.conf.local /etc/bind/named.conf.local
=> [5/5] COPY db.csne.vcct.com /etc/bind/db.csne.vcct.com
=> => exporting to image
=> => exporting layers
=> => writing image sha256:158510f96bcda5b7c3416827d21c9605a94c96fb74200643a70960483b73e4e
=> => naming to docker.io/it22310132/dns-server:1.0
thinukam@thinukam:~/vcct/dns-server$

```

Figure 2. 6 - Build the docker image for DNS server

```
thinukam@thinukam: ~/vcct/dns-server
thinukam@thinukam:~/vcct/dns-server$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
<none>              <none>     158510f96bcd About a minute ago 156MB
it22310132/dns-server 1.0        c67f89cc6375 About a minute ago 156MB
hello-world         latest     ca9905c726f0 7 weeks ago    5.2kB
thinukam@thinukam:~/vcct/dns-server$
```

Figure 2. 7 - Verify the docker image for DNS server

- Deploy the docker image as a container using the **docker run** command, and verify the status of the service with **docker ps** command.

```
thinukam@thinukam: ~/vcct/dns-server
thinukam@thinukam:~/vcct/dns-server$ docker run -d --name dns-container --network host --restart always it22310132/dns-server:1.0
2aa4186b0937c67d4a2e42d682b02a99ed263a6774c5d1c0bf76a67ecb38ca8f
thinukam@thinukam:~/vcct/dns-server$
thinukam@thinukam:~/vcct/dns-server$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS   NAMES
2aa4186b0937   it22310132/dns-server:1.0 "named -g -c /etc/bi..." 25 seconds ago Up 24 seconds      dns-container
thinukam@thinukam:~/vcct/dns-server$
```

Figure 2. 8 - Deploy the docker image for DNS server and verify status

- On different host, test resolutions for *csne.vcct.com* and *google.com* using **dig**.

```
thinukadm@fedora:~
thinukadm@fedora:~$ dig @192.168.8.100 csne.vcct.com

; <<>> DiG 9.18.33 <<>> @192.168.8.100 csne.vcct.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25036
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: e6a708173a54063b0100000068dd7b7b66301b8d02fda3b1 (good)
;; QUESTION SECTION:
;csne.vcct.com.                IN      A

;; ANSWER SECTION:
csne.vcct.com.                604800  IN      A      192.168.8.132

;; Query time: 0 msec
;; SERVER: 192.168.8.100#53(192.168.8.100) (UDP)
;; WHEN: Thu Oct 02 06:05:21 +0530 2025
;; MSG SIZE rcvd: 86

thinukadm@fedora:~$
```

Figure 2. 9 - DNS resolution test case 1

```
thinukadm@fedora:~$ dig @192.168.8.100 google.com

; <<>> DiG 9.18.33 <<>> @192.168.8.100 google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 19495
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 98cd149744aa37110100000068dd7ba2c44c507399b1c16b (good)
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                142     IN      A      172.217.194.139
google.com.                142     IN      A      172.217.194.138
google.com.                142     IN      A      172.217.194.113
google.com.                142     IN      A      172.217.194.100
google.com.                142     IN      A      172.217.194.102
google.com.                142     IN      A      172.217.194.101

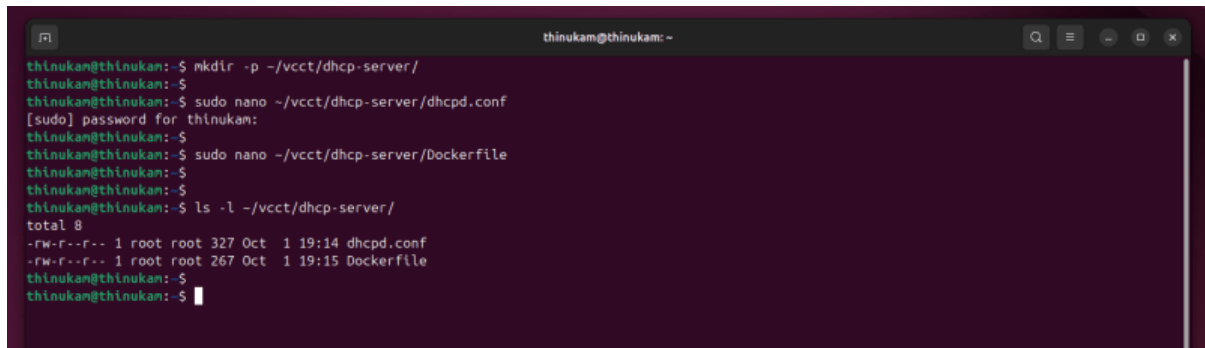
;; Query time: 1 msec
;; SERVER: 192.168.8.100#53(192.168.8.100) (UDP)
;; WHEN: Thu Oct 02 06:05:57 +0530 2025
;; MSG SIZE rcvd: 163

thinukadm@fedora:~$
```

Figure 2.10 - DNS resolution test case 2

3. DHCP Server

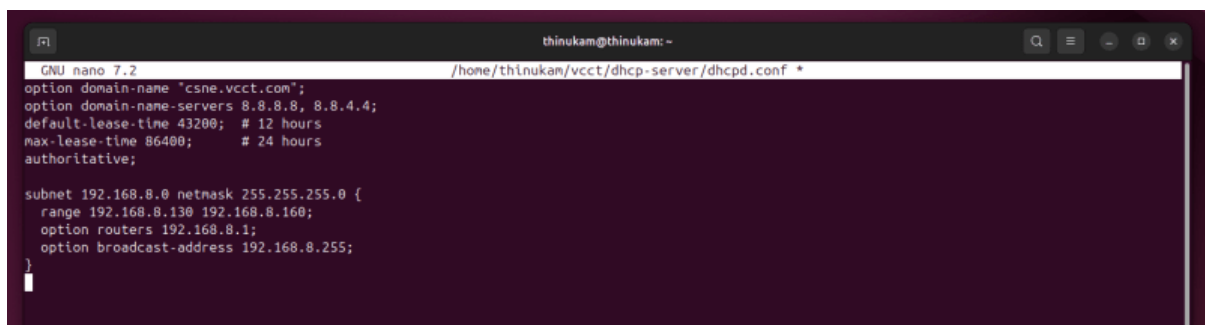
- Several files were maintained under the following directory hierarchy to create the Docker image for DHCP server.



```
thinukam@thinukam:~$ mkdir -p ~/vcct/dhcp-server/
thinukam@thinukam:~$ 
thinukam@thinukam:~$ sudo nano ~/vcct/dhcp-server/dhcpd.conf
[sudo] password for thinukam:
thinukam@thinukam:~$ 
thinukam@thinukam:~$ sudo nano ~/vcct/dhcp-server/Dockerfile
thinukam@thinukam:~$ 
thinukam@thinukam:~$ 
thinukam@thinukam:~$ ls -l ~/vcct/dhcp-server/
total 8
-rw-r--r-- 1 root root 327 Oct  1 19:14 dhcpd.conf
-rw-r--r-- 1 root root 267 Oct  1 19:15 Dockerfile
thinukam@thinukam:~$ 
thinukam@thinukam:~$
```

Figure 3. 1 - Directory hierarchy for DHCP

- The *dhcpd.conf* configuration file was created to define the IP address pool, lease time, default gateway, and DNS server options.



```
GNU nano 7.2 /home/thinukam/vcct/dhcp-server/dhcpd.conf *
option domain-name "csne.vcct.com";
option domain-name-servers 8.8.8.8, 8.8.4.4;
default-lease-time 43200; # 12 hours
max-lease-time 86400; # 24 hours
authoritative;

subnet 192.168.8.0 netmask 255.255.255.0 {
    range 192.168.8.130 192.168.8.160;
    option routers 192.168.8.1;
    option broadcast-address 192.168.8.255;
}
```

Figure 3. 2 - dhcpd.conf configuration file

- The Dockerfile was created to automate the setup of the DHCP server by installing required *isc-dhcp-server* packages and copying the configuration file into the container.



```
GNU nano 7.2 /home/thinukam/vcct/dhcp-server/Dockerfile *
FROM ubuntu:22.04

RUN apt-get update && \
    apt-get install -y isc-dhcp-server && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/* && \
    mkdir -p /var/lib/dhcp && \
    touch /var/lib/dhcp/dhcpd.leases && \
    chown dhcpd:dhcpd /var/lib/dhcp/dhcpd.leases

COPY dhcpd.conf /etc/dhcp/dhcpd.conf

ENTRYPOINT ["/usr/sbin/dhcpd", "-f", "-d", "--no-pid", "-cf", "/etc/dhcp/dhcpd.conf"]
```

Figure 3. 3 - Dockerfile to automate DHCP setup

- Change the current working directory to the directory where the DHCP Dockerfile is maintained (~/*vcct/dhcp-server*/ in this instance).
- Build the docker image for the DHCP server and verify the successful creation of the image.

```
thinukam@thinukam: ~/vcct/dhcp-server
thinukam@thinukam:~/vcct/dhcp-server$ docker build -t it22310132/dhcp-server:1.0 .
[+] Building 15.5s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 435B
=> [internal] load metadata for docker.io/library/ubuntu:22.04
=> [auth] library/ubuntu:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/3] FROM docker.io/library/ubuntu:22.04@sha256:4e0171b9275e12d375863f2b3ae9ce00a4c53ddda176bd55868df97ac6f21a6e
=> [internal] load build context
=> => transferring context: 32B
=> [2/3] RUN apt-get update && apt-get install -y isc-dhcp-server && apt-get clean && rm -rf /var/lib/apt/lists/* && mk
=> [3/3] COPY dhcpd.conf /etc/dhcp/dhcpd.conf
=> exporting to image
=> => exporting layers
=> => writing image sha256:5a2365e8382bcbcd0d289283f651046837d1116e4d52402734e922831b0470
=> => naming to docker.io/it22310132/dhcp-server:1.0
thinukam@thinukam:~/vcct/dhcp-server$
thinukam@thinukam:~/vcct/dhcp-server$
thinukam@thinukam:~/vcct/dhcp-server$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
it22310132/dhcp-server 1.0                5a2365e8382b       6 seconds ago      74.5MB
<none>               <none>             2baf9827b425       7 minutes ago      74.5MB
<none>               <none>             158510f96bcd       28 minutes ago     156MB
it22310132/dns-server 1.0                c67f89cc6375       28 minutes ago     156MB
hello-world          latest             ca9905c726f0       7 weeks ago        5.2kB
thinukam@thinukam:~/vcct/dhcp-server$
```

Figure 3. 4 - Build the Docker image for DHCP server

- Deploy the docker image as a container using the ***docker run*** command, and verify the status of the service with ***docker ps*** command.

```
thinukam@thinukam: ~/vcct/dhcp-server
thinukam@thinukam:~/vcct/dhcp-server$ docker run -d --name dhcp-container --network host --cap-add=NET_ADMIN --cap-add=NET_RAW --restart always
it22310132/dhcp-server:1.0
013ec9361c0ef258f6d98a12d3592990c909d98c4a7071c5718d39ba51938d38
thinukam@thinukam:~/vcct/dhcp-server$
thinukam@thinukam:~/vcct/dhcp-server$
thinukam@thinukam:~/vcct/dhcp-server$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS   NAMES
013ec9361c0e   it22310132/dhcp-server:1.0          "/usr/sbin/dhcpd -f ..." 4 seconds ago Up 4 seconds          dhcp-container
2aa4106b0937   it22310132/dns-server:1.0          "named -g -c /etc/bi..." 27 minutes ago Up 27 minutes          dns-container
thinukam@thinukam:~/vcct/dhcp-server$
```

Figure 3. 5 - Deploy the docker image for DHCP server and verify status

- Check the docker logs for dhcp to verify the correct interface is listening and sending DHCP information via the correct interface.

```

thinukam@thinukam: ~/vcct/dhcp-server
thinukam@thinukam:~/vcct/dhcp-server$ docker run -d --name dhcp-container --network host --cap-add=NET_ADMIN --cap-add=NET_RAW --restart always
it22310132/dhcp-server:1.0
013ec9361c0ef258f6d98a12d3592990c909d90c4a7071c5718d39ba51938d38
thinukam@thinukam:~/vcct/dhcp-server$
thinukam@thinukam:~/vcct/dhcp-server$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
013ec9361c0e   it22310132/dhcp-server:1.0         "/usr/sbin/dhcpd -f ..." 4 seconds ago  Up 4 seconds  0.0.0.0:0.0.0  dhcp-container
2aa4186b0937   it22310132/dns-server:1.0         "named -g -c /etc/bi..." 27 minutes ago Up 27 minutes  0.0.0.0:0.0.0  dns-container
thinukam@thinukam:~/vcct/dhcp-server$ docker logs dhcp-container
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Wrote 0 leases to leases file.

No subnet declaration for docker0 (172.17.0.1).
** Ignoring requests on docker0. If this is not what
you want, please write a subnet declaration
in your dhcpd.conf file for the network segment
to which interface docker0 is attached. **

No subnet declaration for ens256 (192.168.231.168).
** Ignoring requests on ens256. If this is not what
you want, please write a subnet declaration
in your dhcpd.conf file for the network segment
to which interface ens256 is attached. **

Listening on LPF/ens160/00:0c:29:fe:a0:4c/192.168.8.0/24
Sending on   LPF/ens160/00:0c:29:fe:a0:4c/192.168.8.0/24
Sending on   Socket/fallback/fallback-net
Server starting service.
thinukam@thinukam:~/vcct/dhcp-server$

```

Figure 3. 6 – Docker logs of dhcp container (i)

- On the client machine, release the current DHCP information and renew the DHCP information by sending a request.

```

thinukadm@fedora:~$ sudo dhclient -v -r
[sudo] password for thinukadm:
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

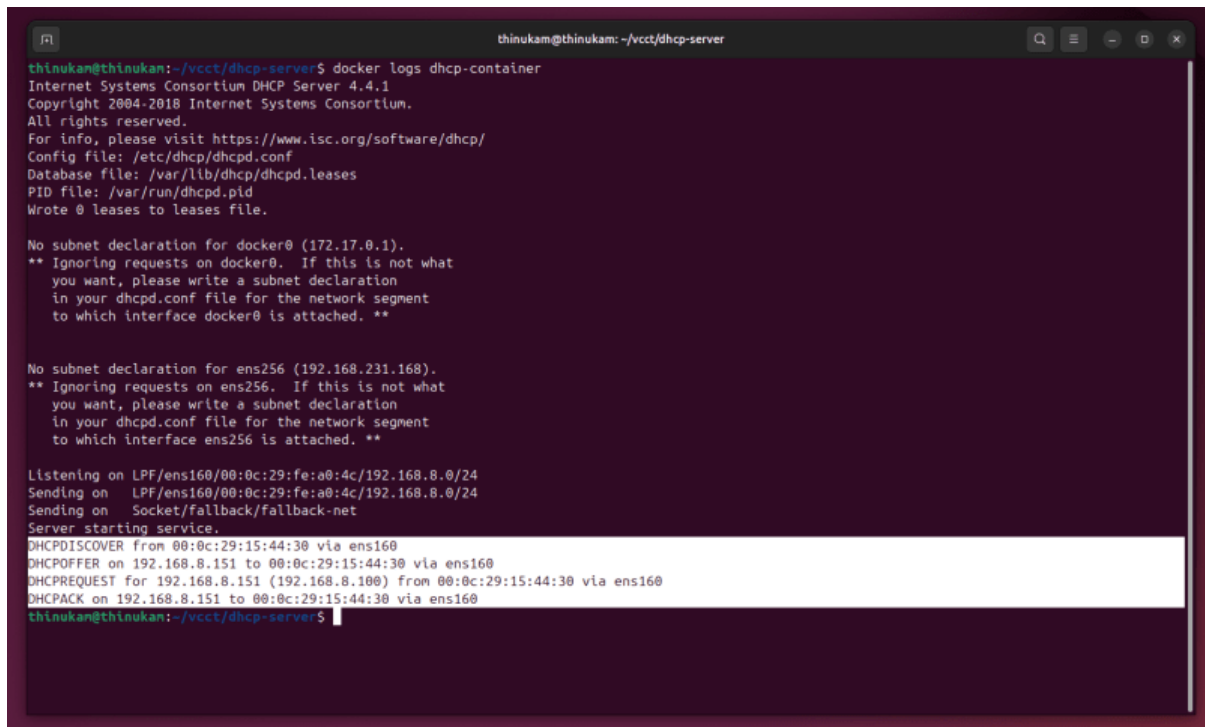
Listening on LPF/ens160/00:0c:29:15:44:30
Sending on   LPF/ens160/00:0c:29:15:44:30
Sending on   Socket/fallback
DHCPRELEASE of 192.168.8.151 on ens160 to 192.168.8.50 port 67 (xid=0xbd639f3f)
thinukadm@fedora:~$
thinukadm@fedora:~$ sudo dhclient -v
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/ens160/00:0c:29:15:44:30
Sending on   LPF/ens160/00:0c:29:15:44:30
Sending on   Socket/fallback
DHCPDISCOVER on ens160 to 255.255.255.255 port 67 interval 5 (xid=0xe80c1e18)
DHCPOFFER of 192.168.8.151 from 192.168.8.100
DHCPREQUEST for 192.168.8.151 on ens160 to 255.255.255.255 port 67 (xid=0xe80c1e18)
DHCPACK of 192.168.8.151 from 192.168.8.100 (xid=0xe80c1e18)
/usr/sbin/dhclient-script: line 706: /etc/resolv.conf: Permission denied
bound to 192.168.8.151 -- renewal in 17620 seconds.
thinukadm@fedora:~$

```

Figure 3. 7 - Release and renew DHCP information on client

- Check the docker logs of the dhcp container and verify DHCPDISCOVER, DHCPOFFER, DHCPREQUEST, and DHCPACK messages are visible.



```
thinukam@thinukam: ~/vcct/dhcp-server
thinukam@thinukam:~/vcct/dhcp-server$ docker logs dhcp-container
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Wrote 0 leases to leases file.

No subnet declaration for docker0 (172.17.0.1).
** Ignoring requests on docker0. If this is not what
you want, please write a subnet declaration
in your dhcpd.conf file for the network segment
to which interface docker0 is attached. **

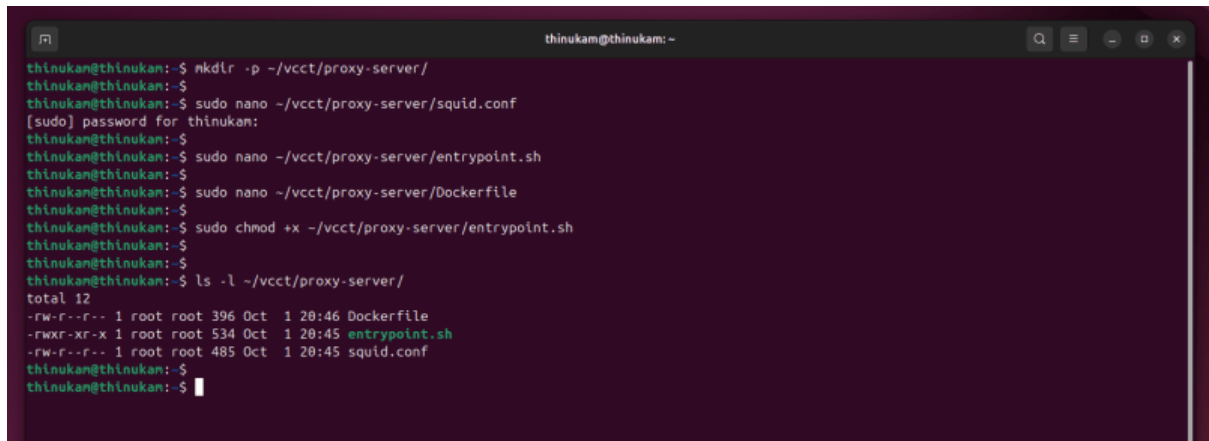
No subnet declaration for ens256 (192.168.231.168).
** Ignoring requests on ens256. If this is not what
you want, please write a subnet declaration
in your dhcpd.conf file for the network segment
to which interface ens256 is attached. **

Listening on LPF/ens160/00:0c:29:fe:a0:4c/192.168.8.0/24
Sending on   LPF/ens160/00:0c:29:fe:a0:4c/192.168.8.0/24
Sending on   Socket/fallback/fallback-net
Server starting service.
DHCPDISCOVER from 00:0c:29:15:44:30 via ens160
DHCPOFFER on 192.168.8.151 to 00:0c:29:15:44:30 via ens160
DHCPREQUEST for 192.168.8.151 (192.168.8.100) from 00:0c:29:15:44:30 via ens160
DHCPACK on 192.168.8.151 to 00:0c:29:15:44:30 via ens160
thinukam@thinukam:~/vcct/dhcp-server$
```

Figure 3. 8 - Docker logs of dhcp container (ii)

4. Proxy Server

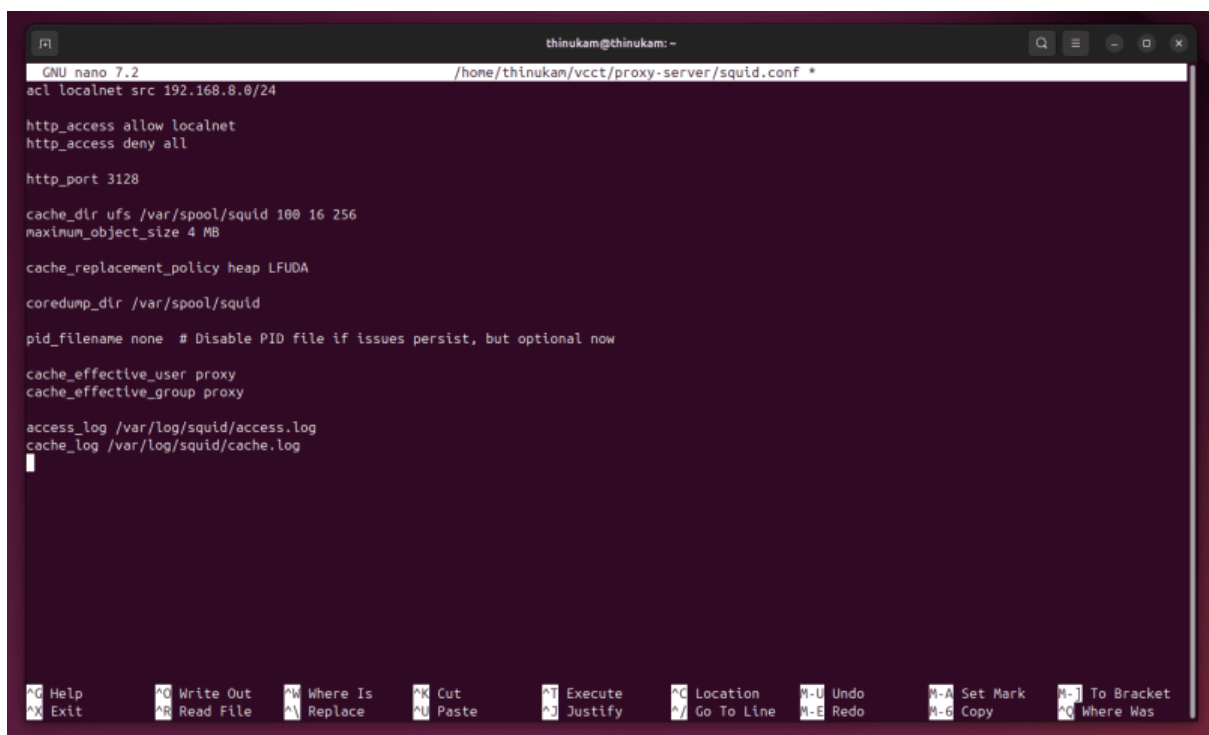
- Several files were maintained under the following directory hierarchy to create the docker image for Squid proxy server.



```
thinukam@thinukam: ~$ mkdir -p ~/vcct/proxy-server/
thinukam@thinukam: ~$ 
thinukam@thinukam: ~$ sudo nano ~/vcct/proxy-server/squid.conf
[sudo] password for thinukam:
thinukam@thinukam: ~$ 
thinukam@thinukam: ~$ sudo nano ~/vcct/proxy-server/entrypoint.sh
thinukam@thinukam: ~$ 
thinukam@thinukam: ~$ sudo nano ~/vcct/proxy-server/Dockerfile
thinukam@thinukam: ~$ 
thinukam@thinukam: ~$ sudo chmod +x ~/vcct/proxy-server/entrypoint.sh
thinukam@thinukam: ~$ 
thinukam@thinukam: ~$ ls -l ~/vcct/proxy-server/
total 12
-rw-r--r-- 1 root root 396 Oct 1 20:46 Dockerfile
-rwxr-xr-x 1 root root 534 Oct 1 20:45 entrypoint.sh
-rw-r--r-- 1 root root 485 Oct 1 20:45 squid.conf
thinukam@thinukam: ~$ 
thinukam@thinukam: ~$
```

Figure 4. 1 - Directory hierarchy for Proxy

- The **squid.conf** configuration file defines the proxy rules, including access control lists, listening port, cache size, maximum object size, etc.



```
GNU nano 7.2 /home/thinukam/vcct/proxy-server/squid.conf *
acl localnet src 192.168.0.0/24

http_access allow localnet
http_access deny all

http_port 3128

cache_dir ufs /var/spool/squid 100 16 256
maximum_object_size 4 MB

cache_replacement_policy heap LFUDA

coredump_dir /var/spool/squid

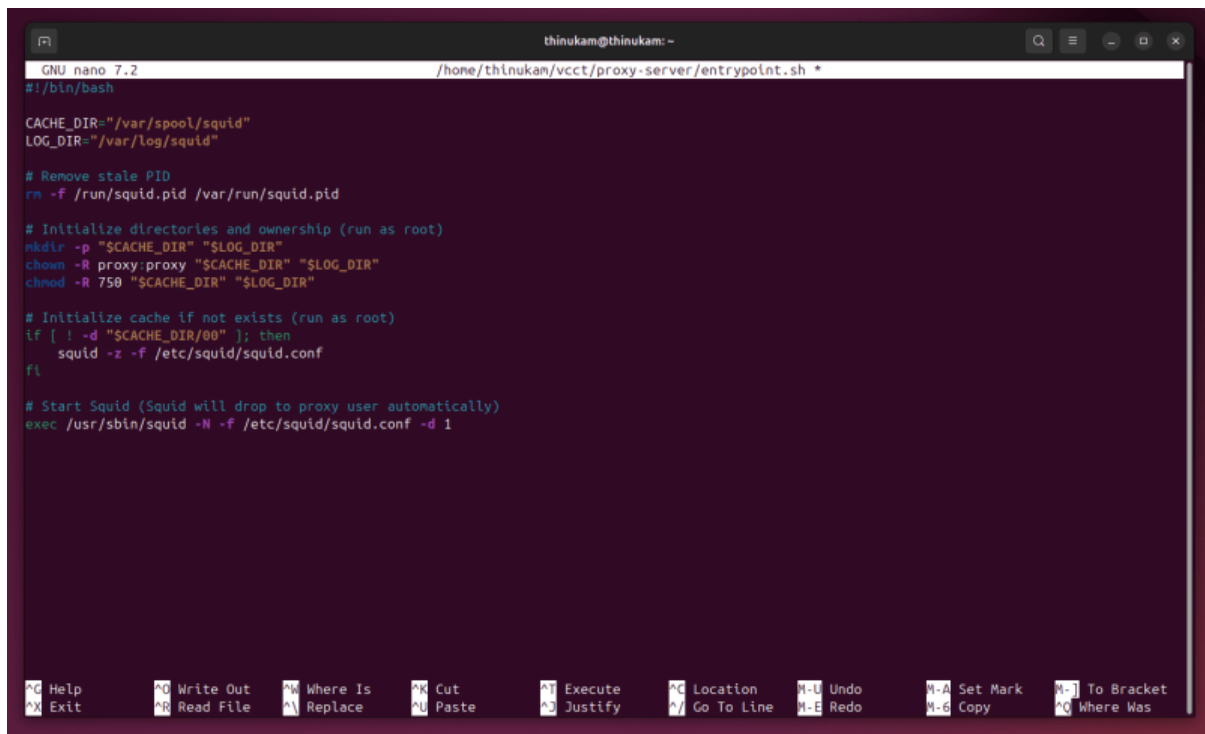
pid_filename none # Disable PID file if issues persist, but optional now

cache_effective_user proxy
cache_effective_group proxy

access_log /var/log/squid/access.log
cache_log /var/log/squid/cache.log
^H^H
```

Figure 4. 2 - squid.conf configuration file

- The **entrypoint.sh** script was developed to automate container startup tasks. This script clears stale PID files, creates and assigns appropriate ownership to cache and log directories, initializes the Squid cache if it has not been created, and finally starts the Squid process in the foreground. This ensures that the container launches consistently, even after restarts.
- Ensure to change the file permission to executable.



```

thinukam@thinukam: ~
GNU nano 7.2 /home/thinukam/vcct/proxy-server/entrypoint.sh *
#!/bin/bash

CACHE_DIR="/var/spool/squid"
LOG_DIR="/var/log/squid"

# Remove stale PID
rm -f /run/squid.pid /var/run/squid.pid

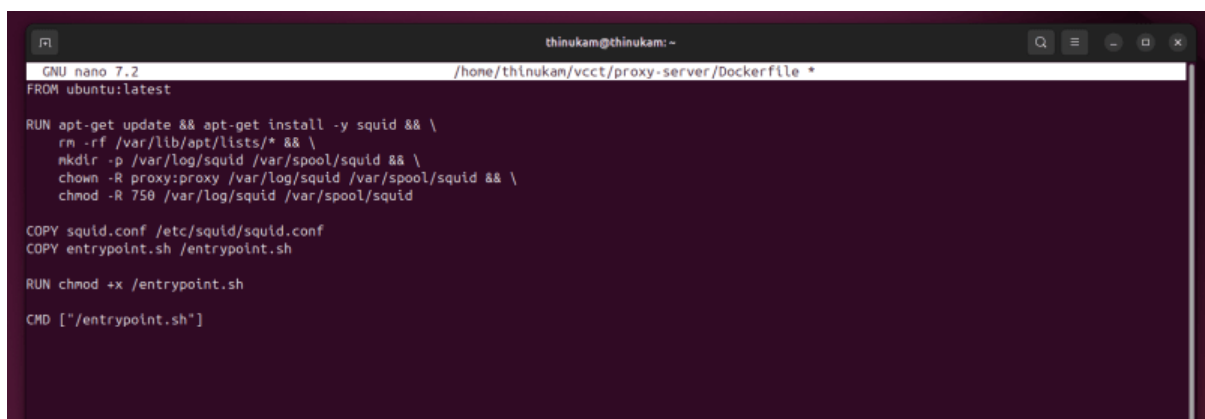
# Initialize directories and ownership (run as root)
mkdir -p "$CACHE_DIR" "$LOG_DIR"
chown -R proxy:proxy "$CACHE_DIR" "$LOG_DIR"
chmod -R 750 "$CACHE_DIR" "$LOG_DIR"

# Initialize cache if not exists (run as root)
if [ ! -d "$CACHE_DIR/00" ]; then
    squid -z -f /etc/squid/squid.conf
fi

# Start Squid (Squid will drop to proxy user automatically)
exec /usr/sbin/squid -N -f /etc/squid/squid.conf -d 1
  
```

Figure 4. 3 - *entrypoint.sh* script file

- The Dockerfile was created to automate the setup of the Proxy server by installing the Squid package, preparing the necessary directories, copying both the configuration file and the script file into the container, and so on.



```

thinukam@thinukam: ~
GNU nano 7.2 /home/thinukam/vcct/proxy-server/Dockerfile *
FROM ubuntu:latest

RUN apt-get update && apt-get install -y squid && \
    rm -rf /var/lib/apt/lists/* && \
    mkdir -p /var/log/squid /var/spool/squid && \
    chown -R proxy:proxy /var/log/squid /var/spool/squid && \
    chmod -R 750 /var/log/squid /var/spool/squid

COPY squid.conf /etc/squid/squid.conf
COPY entrypoint.sh /entrypoint.sh

RUN chmod +x /entrypoint.sh

CMD ["/entrypoint.sh"]
  
```

Figure 4. 4 - *Dockerfile to automate Squid setup*

- Change the current working directory to the directory where the Squid Dockerfile is maintained (~/.vcct/proxy-server/ in this instance).
- Build the docker image for the Proxy server and verify the successful creation of the image.

```
thinukam@thinukam: ~/.vcct/proxy-server/
thinukam@thinukam: ~/.vcct/proxy-server$
thinukam@thinukam: ~/.vcct/proxy-server$ docker build -t it22310132/proxy-server:1.0 .
[+] Building 29.5s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 435B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [auth] library/ubuntu:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/5] FROM docker.io/library/ubuntu:latest@sha256:353675e2a41badd526e2b837d7ec780c2a05bca0164f7ea5dbbd433d21d166fc
=> [internal] load build context
=> => transferring context: 1.10kB
=> [2/5] RUN apt-get update && apt-get install -y squid && rm -rf /var/lib/apt/lists/* && mkdir -p /var/log/squid /var/spool/sq
=> [3/5] COPY squid.conf /etc/squid/squid.conf
=> [4/5] COPY entrypoint.sh /entrypoint.sh
=> [5/5] RUN chmod +x /entrypoint.sh
=> exporting to image
=> => exporting layers
=> => writing image sha256:adffcccf170a7916e0351dcd4c388db0dcd57316319709c71ca7ed5bfa63442f
=> => naming to docker.io/it22310132/proxy-server:1.0
thinukam@thinukam: ~/.vcct/proxy-server$
thinukam@thinukam: ~/.vcct/proxy-server$
thinukam@thinukam: ~/.vcct/proxy-server$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
it22310132/proxy-server	1.0	adffcccf170a	11 seconds ago	296MB
it22310132/dhcp-server	1.0	5a2365e8382b	About an hour ago	74.5MB
<none>	<none>	2ba79827b425	2 hours ago	74.5MB
<none>	<none>	158510f96bcd	2 hours ago	156MB
it22310132/dns-server	1.0	c67f89cc6375	2 hours ago	156MB
hello-world	latest	ca9905c726f0	7 weeks ago	5.2kB

```
thinukam@thinukam: ~/.vcct/proxy-server$
```

Figure 4. 5 - Build the Docker image for Proxy server

- Deploy the docker image as a container using the **docker run** command, and verify the status of the service with **docker ps** command.

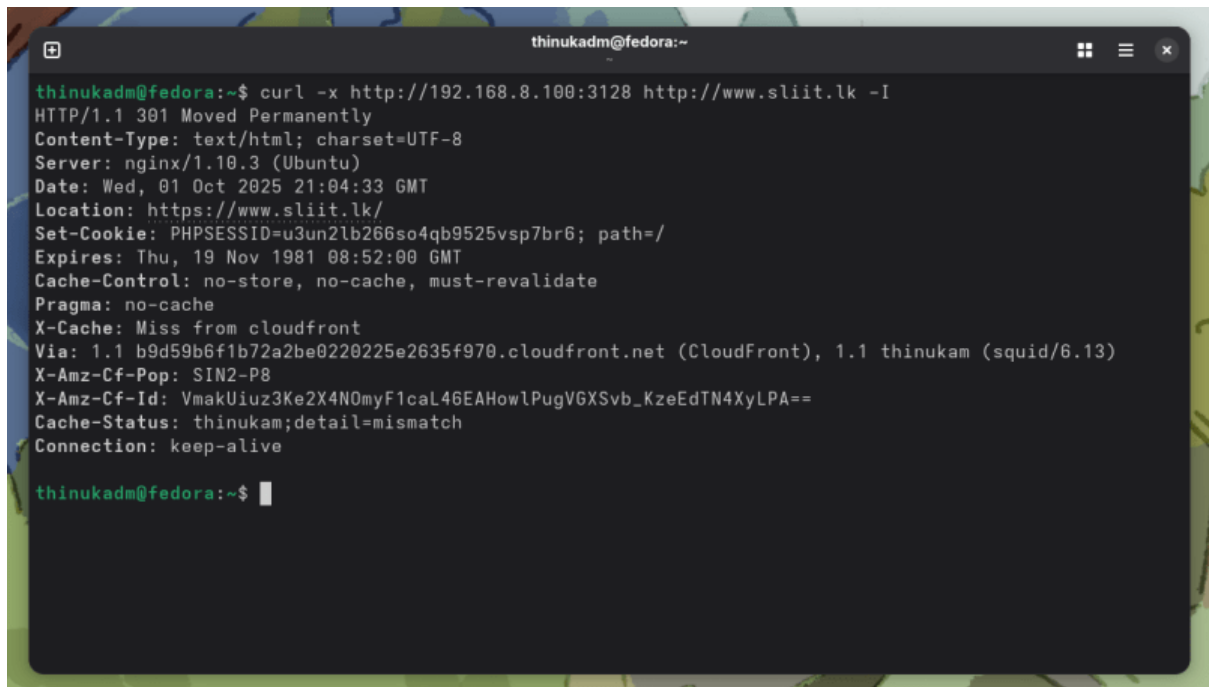
```
thinukam@thinukam: ~/.vcct/proxy-server/
thinukam@thinukam: ~/.vcct/proxy-server$
thinukam@thinukam: ~/.vcct/proxy-server$ docker run -d --name proxy-container --network host -p 3128:3128 --cpus=1 --memory=512m --restart always
it22310132/proxy-server:1.0
WARNING: Published ports are discarded when using host network mode
13abb917ca7023ff662d67425441c8811e2ec3f08779a039e0e89f1971711dd1
thinukam@thinukam: ~/.vcct/proxy-server$
thinukam@thinukam: ~/.vcct/proxy-server$
thinukam@thinukam: ~/.vcct/proxy-server$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
13abb917ca70	it22310132/proxy-server:1.0	"/entrypoint.sh"	3 seconds ago	Up 3 seconds		proxy-container
013ec9361c0e	it22310132/dhcp-server:1.0	"/usr/sbin/dhcpd -f ..."	About an hour ago	Up About an hour		dhcp-container
2aa4186b0937	it22310132/dns-server:1.0	"named -g -c /etc/bl..."	2 hours ago	Up 2 hours		dns-container

```
thinukam@thinukam: ~/.vcct/proxy-server$
```

Figure 4. 6 - Deploy the docker image for Proxy server and verify status

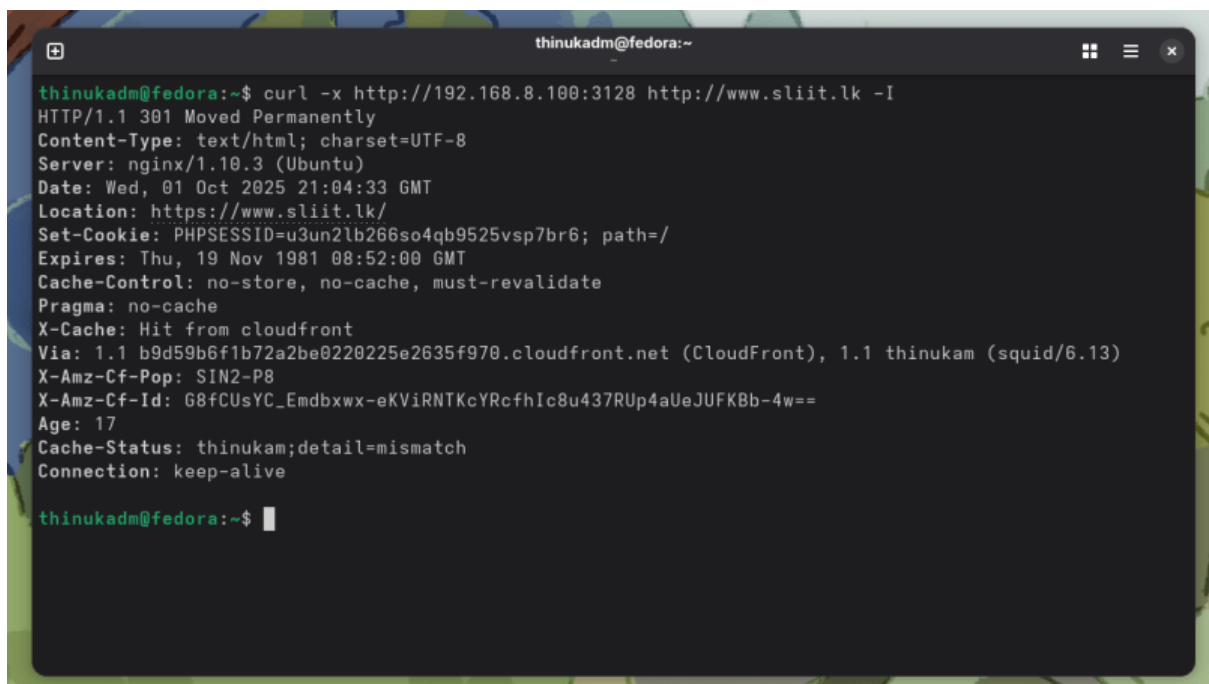
- On the client machine access sites via proxy using *curl*. Run twice each, as the second time should be faster indicating a cache hit.



```
thinukadm@fedora:~$ curl -x http://192.168.8.100:3128 http://www.sliit.lk -I
HTTP/1.1 301 Moved Permanently
Content-Type: text/html; charset=UTF-8
Server: nginx/1.10.3 (Ubuntu)
Date: Wed, 01 Oct 2025 21:04:33 GMT
Location: https://www.sliit.lk/
Set-Cookie: PHPSESSID=u3un2lb266so4qb9525vsp7br6; path=/
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
X-Cache: Miss from cloudfront
Via: 1.1 b9d59b6f1b72a2be0220225e2635f970.cloudfront.net (CloudFront), 1.1 thinukam (squid/6.13)
X-Amz-Cf-Pop: SIN2-P8
X-Amz-Cf-Id: VmakUiuz3Ke2X4N0myF1caL46EAHowlPugVGXSvb_KzeEdTN4XyLPA==
Cache-Status: thinukam;detail=mismatch
Connection: keep-alive

thinukadm@fedora:~$
```

Figure 4. 7 - curl test on client (i)



```
thinukadm@fedora:~$ curl -x http://192.168.8.100:3128 http://www.sliit.lk -I
HTTP/1.1 301 Moved Permanently
Content-Type: text/html; charset=UTF-8
Server: nginx/1.10.3 (Ubuntu)
Date: Wed, 01 Oct 2025 21:04:33 GMT
Location: https://www.sliit.lk/
Set-Cookie: PHPSESSID=u3un2lb266so4qb9525vsp7br6; path=/
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
X-Cache: Hit from cloudfront
Via: 1.1 b9d59b6f1b72a2be0220225e2635f970.cloudfront.net (CloudFront), 1.1 thinukam (squid/6.13)
X-Amz-Cf-Pop: SIN2-P8
X-Amz-Cf-Id: G8fCUUsYC_Emdbxwx-eKViRNTKcYRcfhIc8u437RUp4aUeJUFKBb-4w==
Age: 17
Cache-Status: thinukam;detail=mismatch
Connection: keep-alive

thinukadm@fedora:~$
```

Figure 4. 8 - curl test on client (ii)

- The following figure is the log of cached proxy data, and the last 2 records indicate the website (<http://httpforever.com/>) was accessed twice.
- Initially a **TCP_MISS** indicating a cache miss, and then a **TCP_MEM_HIT** indicating a cache hit.

```

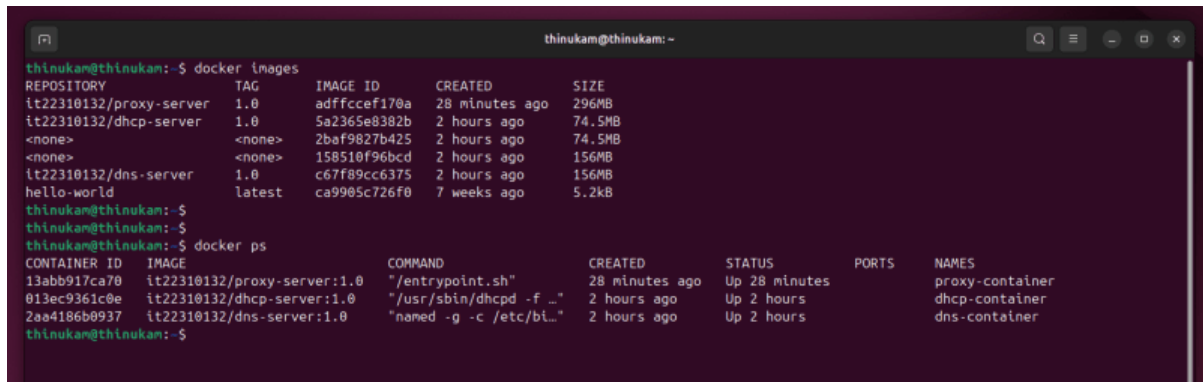
thinukam@thinukam:~$ docker exec -it proxy-container cat /var/log/squid/access.log
1759351966.344 191 192.168.8.130 TCP_MISS/200 18803 GET http://www.google.com/ - HIER_DIRECT/172.253.118.99 text/html
1759352020.268 0 127.0.0.1 TCP_DENIED/403 3809 GET http://google.com/ - HIER_NONE/- text/html
1759352246.649 111 192.168.8.130 TCP_MISS/200 1119 HEAD http://www.google.com/ - HIER_DIRECT/172.253.118.147 text/html
1759352296.458 52 192.168.8.130 TCP_MISS/200 1119 HEAD http://www.google.com/ - HIER_DIRECT/172.253.118.147 text/html
1759352355.907 783 192.168.8.130 TCP_MISS/301 630 HEAD http://www.sliit.lk/ - HIER_DIRECT/13.35.238.21 text/html
1759352359.496 45 192.168.8.130 TCP_MISS/301 637 HEAD http://www.sliit.lk/ - HIER_DIRECT/13.35.238.21 text/html
1759352477.274 105 192.168.8.130 TCP_MISS/301 473 HEAD http://www.youtube.com/ - HIER_DIRECT/142.251.12.91 application/binary
1759352490.320 129 192.168.8.130 TCP_MISS/301 469 HEAD http://youtube.com/ - HIER_DIRECT/74.125.68.93 application/binary
1759352562.971 0 127.0.0.1 TCP_DENIED/403 3809 GET http://google.com/ - HIER_NONE/- text/html
1759352583.467 108 192.168.8.100 TCP_MISS/301 869 GET http://google.com/ - HIER_DIRECT/74.125.200.138 text/html
1759352597.517 166 192.168.8.100 TCP_MISS/200 18812 GET http://www.google.com/ - HIER_DIRECT/142.251.12.104 text/html
1759352657.201 134 192.168.8.130 TCP_MISS/301 393 HEAD http://draw.io/ - HIER_DIRECT/104.18.9.17 text/html
1759352673.918 803 192.168.8.130 TCP_MISS/301 630 HEAD http://www.sliit.lk/ - HIER_DIRECT/13.35.238.6 text/html
1759352690.931 54 192.168.8.130 TCP_MISS/301 630 HEAD http://www.sliit.lk/ - HIER_DIRECT/13.35.238.6 text/html
1759352714.656 32 192.168.8.130 TCP_MISS_ABORTED/502 345 HEAD http://courseweb.sliit.lk/ - HIER_DIRECT/122.255.11.220 text/html
1759352721.462 1070 192.168.8.130 TCP_MISS_ABORTED/503 345 HEAD http://www.courseweb.sliit.lk/ - HIER_NONE/- text/html
1759352741.664 167 192.168.8.130 TCP_MISS/301 473 HEAD http://www.youtube.com/ - HIER_DIRECT/64.233.170.93 application/binary
1759352758.619 304 192.168.8.130 TCP_TUNNEL/200 9634 CONNECT www.youtube.com:443 - HIER_DIRECT/64.233.170.93 -
1759352787.847 106 192.168.8.130 TCP_MISS/200 1119 HEAD http://www.google.com/ - HIER_DIRECT/142.251.12.103 text/html
1759353028.497 543 192.168.8.130 TCP_MISS/200 313 HEAD http://example.com/ - HIER_DIRECT/23.220.75.245 text/html
1759353036.615 0 192.168.8.130 TCP_MEM_HIT/200 322 HEAD http://example.com/ - HIER_NONE/- text/html
1759353221.510 1222 192.168.8.130 TCP_MISS_ABORTED/503 345 HEAD http://csne.vcct.com/ - HIER_NONE/- text/html
1759353225.249 0 192.168.8.130 TCP_MISS_ABORTED/503 345 HEAD http://csne.vcct.com/ - HIER_NONE/- text/html
1759353769.980 717 192.168.8.130 TCP_MISS/200 896 HEAD http://httpforever.com/ - HIER_DIRECT/146.190.62.39 text/html
1759353785.708 0 192.168.8.130 TCP_MEM_HIT/200 906 HEAD http://httpforever.com/ - HIER_NONE/- text/html
thinukam@thinukam:~$

```

Figure 4. 9 - Cached proxy server data logs

5. Docker Hub

- Before pushing the docker images to docker hub, ensure the correct images are created and verify the status of the containers.

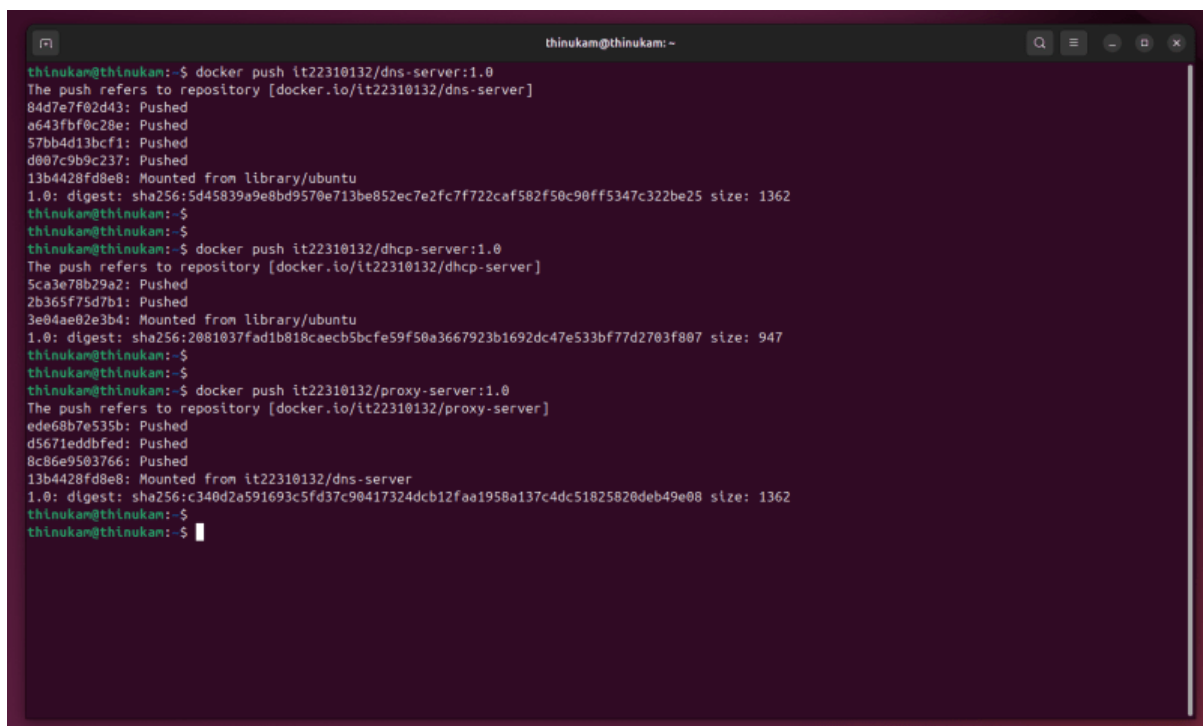


```
thinukam@thinukam:~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
it22310132/proxy-server  1.0                adffcecf170a       28 minutes ago     296MB
it22310132/dhcp-server  1.0                5a2365e8382b       2 hours ago        74.5MB
<none>                <none>             2baf9827b425       2 hours ago        74.5MB
<none>                <none>             158510f96bcd       2 hours ago        156MB
it22310132/dns-server   1.0                c67f89cc6375       2 hours ago        156MB
hello-world            latest             ca9905c726f0       7 weeks ago        5.2kB

thinukam@thinukam:~$ docker ps
CONTAINER ID        IMAGE               COMMAND              CREATED             STATUS              PORTS              NAMES
13abb917ca70       it22310132/proxy-server:1.0  "/entrypoint.sh"     28 minutes ago     Up 28 minutes      0.0.0.0:80->80/tcp  proxy-container
013ec9361c0e       it22310132/dhcp-server:1.0  "/usr/sbin/dhcpd -f ..."  2 hours ago        Up 2 hours          0.0.0.0:67->67/tcp  dhcp-container
2aa4186b0937       it22310132/dns-server:1.0  "named -g -c /etc/bl..."  2 hours ago        Up 2 hours          0.0.0.0:53->53/tcp  dns-container
```

Figure 5. 1 - Verify docker images and container statuses

- Push each docker image to docker hub repository using **docker push** command followed by the docker hub username and the docker image name.



```
thinukam@thinukam:~$ docker push it22310132/dns-server:1.0
The push refers to repository [docker.io/it22310132/dns-server]
84d7e7f02d43: Pushed
a643fbf0c28e: Pushed
57bb4d13bcf1: Pushed
d007c9b9c237: Pushed
13b4428fd8e8: Mounted from library/ubuntu
1.0: digest: sha256:5d45839a9e8bd9570e713be852ec7e2fc7f722caf582f50c90ff5347c322be25 size: 1362
thinukam@thinukam:~$
thinukam@thinukam:~$ docker push it22310132/dhcp-server:1.0
The push refers to repository [docker.io/it22310132/dhcp-server]
5ca3e78b29a2: Pushed
2b365f75d7b1: Pushed
3e04ae02e3b4: Mounted from library/ubuntu
1.0: digest: sha256:2081037fad1b818caecb5bcfe59f50a3667923b1692dc47e533bf77d2703f807 size: 947
thinukam@thinukam:~$
thinukam@thinukam:~$ docker push it22310132/proxy-server:1.0
The push refers to repository [docker.io/it22310132/proxy-server]
ede68b7e535b: Pushed
d5671eddbfed: Pushed
8c86e9503766: Pushed
13b4428fd8e8: Mounted from it22310132/dns-server
1.0: digest: sha256:c340d2a591693c5fd37c90417324dcb12faa1958a137c4dc51825820deb49e08 size: 1362
thinukam@thinukam:~$
thinukam@thinukam:~$
```

Figure 5. 2 - Push docker images to docker hub repository

- Verify the images are appearing on the repository by accessing the docker hub web portal.

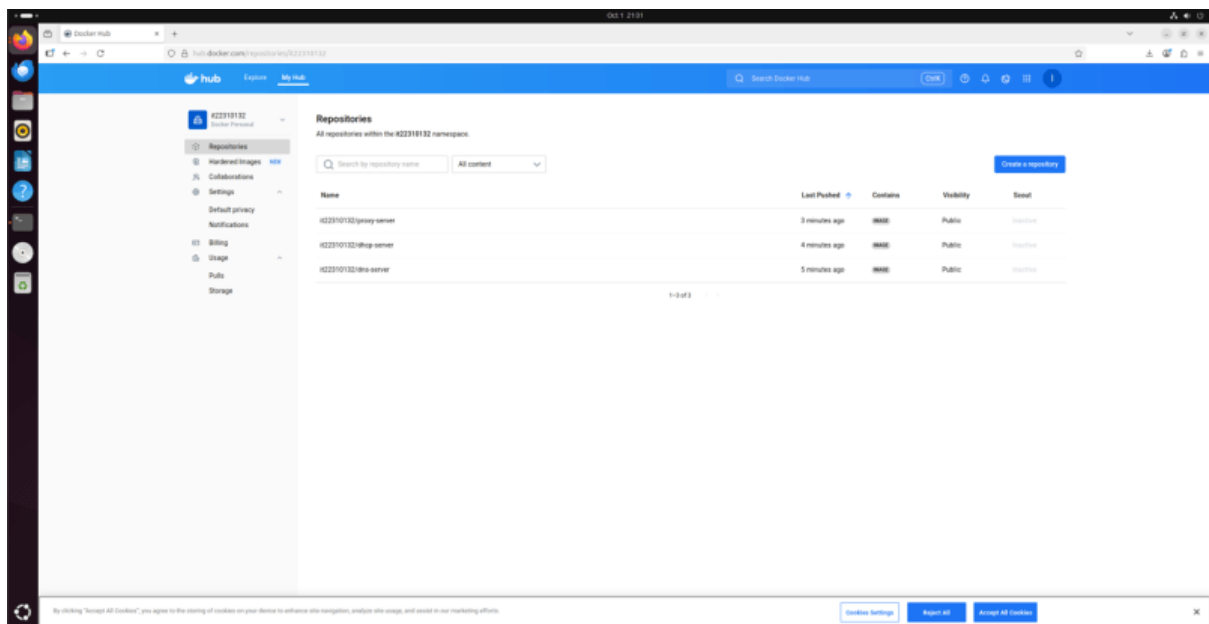


Figure 5. 3 - Verify images in docker hub web portal

- To illustrate the process of pulling Docker images, local copies of Docker images and containers are initially removed.
- To remove the containers, terminate the container processes and subsequently remove the containers.

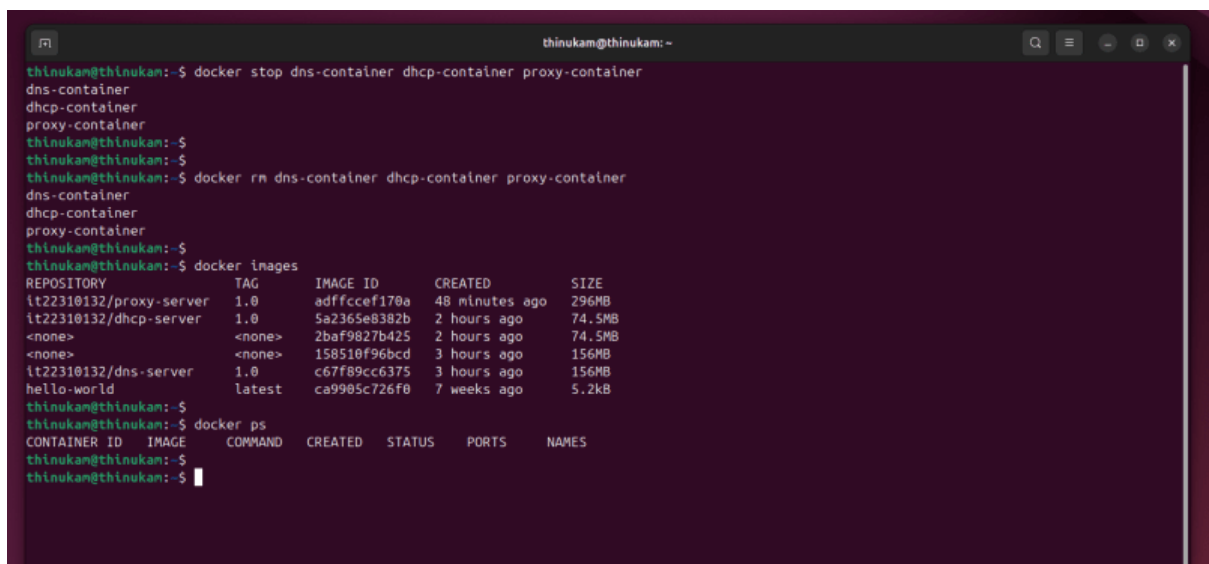


Figure 5. 4 - Terminate and remove the containers

- All local copies of the Docker images must be deleted, and the deletion must be verified.

```
thinukam@thinukam:~$ docker rmi it22310132/dns-server:1.0
Untagged: it22310132/dns-server:1.0
Untagged: it22310132/dns-server@sha256:5d45839a9e8bd9570e713be852ec7e2fc7f722caf582f50c90ff5347c322be25
Deleted: sha256:c67f89cc6375f96fcc8c21b00325902f2376067d316ed9be70573b4f60974502
thinukam@thinukam:~$ docker rmi it22310132/dhcp-server:1.0
Untagged: it22310132/dhcp-server:1.0
Untagged: it22310132/dhcp-server@sha256:2081037fad1b818caecb5bcfe59f50a3667923b1692dc47e533bf77d2703f807
Deleted: sha256:5a2365e8382bcbcd0d289283f651046837d1116e4d52402734e9228331b0470
thinukam@thinukam:~$ docker rmi it22310132/proxy-server:1.0
Untagged: it22310132/proxy-server:1.0
Untagged: it22310132/proxy-server@sha256:c340d2a591693c5fd37c90417324dcb12faa1958a137c4dc51825820deb49e08
Deleted: sha256:adffcccf170a7916e0351dcd4c388db0dcd57316319709c71ca7ed5bfa63442f
thinukam@thinukam:~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
<none>              <none>             2baf9827b425       2 hours ago        74.5MB
<none>              <none>             158510f96bcd       3 hours ago        156MB
hello-world         latest             ca9905c726f0       7 weeks ago        5.2kB
thinukam@thinukam:~$
```

Figure 5. 5 - Remove local copies of docker images

- After completing the previous steps, pull the docker images from the repository using the **docker pull** command followed by the docker hub username and the docker image name.
- Ensure to verify the images are pulled from the docker hub repository.

```
thinukam@thinukam:~$ docker pull it22310132/dns-server:1.0
1.0: Pulling from it22310132/dns-server
59a5d47f84c3: Already exists
f9d00fe43d60: Already exists
0e836f92313b: Already exists
40e85b03b775: Already exists
5c76ab2bec7a: Already exists
Digest: sha256:5d45839a9e8bd9570e713be852ec7e2fc7f722caf582f50c90ff5347c322be25
Status: Downloaded newer image for it22310132/dns-server:1.0
docker.io/it22310132/dns-server:1.0
thinukam@thinukam:~$ docker pull it22310132/dhcp-server:1.0
1.0: Pulling from it22310132/dhcp-server
fdf67ba0bcdc: Already exists
d5528dce15f8: Already exists
abe94af72b38: Already exists
Digest: sha256:2081037fad1b818caecb5bcfe59f50a3667923b1692dc47e533bf77d2703f807
Status: Downloaded newer image for it22310132/dhcp-server:1.0
docker.io/it22310132/dhcp-server:1.0
thinukam@thinukam:~$ docker pull it22310132/proxy-server:1.0
1.0: Pulling from it22310132/proxy-server
59a5d47f84c3: Already exists
3a0f70de2a21: Already exists
2fd5c95bc0dc: Already exists
08549e14e506: Already exists
Digest: sha256:c340d2a591693c5fd37c90417324dcb12faa1958a137c4dc51825820deb49e08
Status: Downloaded newer image for it22310132/proxy-server:1.0
docker.io/it22310132/proxy-server:1.0
thinukam@thinukam:~$
```

Figure 5. 6 - Pull docker images from docker hub repository

```
thinukam@thinukam:~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
it22310132/proxy-server  1.0                adffcccf170a       About an hour ago   296MB
it22310132/dhcp-server  1.0                5a2365e8382b       2 hours ago        74.5MB
<none>              <none>             2baf9827b425       3 hours ago        74.5MB
<none>              <none>             158510f96bcd       3 hours ago        156MB
it22310132/dns-server  1.0                c67f89cc6375       3 hours ago        156MB
hello-world         latest             ca9905c726f0       7 weeks ago        5.2kB
thinukam@thinukam:~$
```

Figure 5. 7 - Verify docker images are pulled

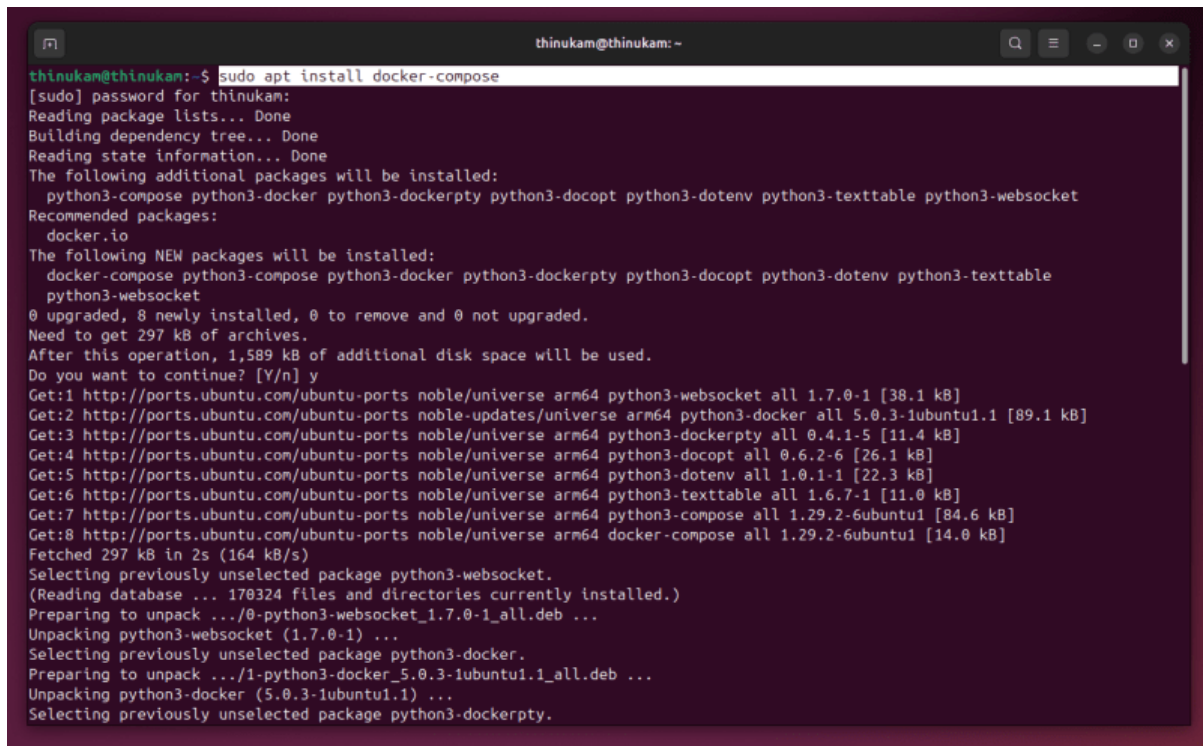
- Subsequently run each docker image as containers with the same set of commands previously used, and verify the status of the containers.

```
thinukam@thinukam:~$ docker run -d --name dns-container --network host --restart always it22310132/dns-server:1.0
9dc2d5b567f3906dd5156baabac0a316408d97c5dcfe860570d075fef627d2a2
thinukam@thinukam:~$ 
thinukam@thinukam:~$ docker run -d --name dhcp-container --network host --cap-add=NET_ADMIN --cap-add=NET_RAW --restart always it22310132/dhcp-server:1.0
419d9d2046688d873f2d65db1e87526141eab34251e25e7991d632ed5216d45a
thinukam@thinukam:~$ 
thinukam@thinukam:~$ docker run -d --name proxy-container --network host -p 3128:3128 --cpus=1 --memory=512m --restart always it22310132/proxy-server:1.0
WARNING: Published ports are discarded when using host network mode
43929d81eada715c93a79d8f9d7a8d9fde09caef68e4ca5c0e605cefb1436e4b
thinukam@thinukam:~$ 
thinukam@thinukam:~$ 
thinukam@thinukam:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
43929d81eada   it22310132/proxy-server:1.0        "/entrypoint.sh"        6 seconds ago Up 5 seconds 3128/tcp     proxy-container
419d9d204668   it22310132/dhcp-server:1.0         "/usr/sbin/dhcpd -f ..." 2 minutes ago Up 2 minutes 68/tcp      dhcp-container
9dc2d5b567f3   it22310132/dns-server:1.0          "named -g -c /etc/bl..." 3 minutes ago  Up 3 minutes 53/tcp      dns-container
thinukam@thinukam:~$
```

Figure 5. 8 - Run docker images and verify

6. Docker Compose File

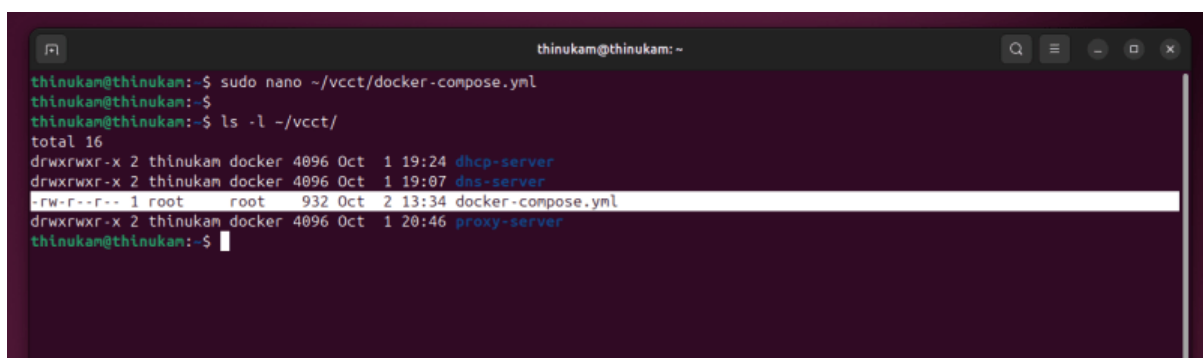
- The docker compose file was created to simultaneously start the containers, rather than subsequently starting each container.
- Before creating the file, **docker-compose** package must be installed to the system

A terminal window titled 'thinukam@thinukam: ~' showing the command 'sudo apt install docker-compose' being executed. The output displays the installation process, including package lists, dependency resolution, and the successful installation of 'docker-compose' along with several other python3-related packages. The terminal text is as follows:

```
thinukam@thinukam:~$ sudo apt install docker-compose
[sudo] password for thinukam:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-compose python3-docker python3-dockerpty python3-dockerpty python3-dockerpty python3-dotenv python3-texttable python3-websocket
Recommended packages:
  docker.io
The following NEW packages will be installed:
  docker-compose python3-compose python3-docker python3-dockerpty python3-dockerpty python3-dotenv python3-texttable python3-websocket
0 upgraded, 8 newly installed, 0 to remove and 0 not upgraded.
Need to get 297 kB of archives.
After this operation, 1,589 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 python3-websocket all 1.7.0-1 [38.1 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 python3-docker all 5.0.3-1ubuntu1.1 [89.1 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 python3-dockerpty all 0.4.1-5 [11.4 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 python3-dockerpty all 0.6.2-6 [26.1 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 python3-dotenv all 1.0.1-1 [22.3 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 python3-texttable all 1.6.7-1 [11.0 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 python3-compose all 1.29.2-6ubuntu1 [84.6 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 docker-compose all 1.29.2-6ubuntu1 [14.0 kB]
Fetched 297 kB in 2s (164 kB/s)
Selecting previously unselected package python3-websocket.
(Reading database ... 170324 files and directories currently installed.)
Preparing to unpack .../0-python3-websocket_1.7.0-1_all.deb ...
Unpacking python3-websocket (1.7.0-1) ...
Selecting previously unselected package python3-docker.
Preparing to unpack .../1-python3-docker_5.0.3-1ubuntu1.1_all.deb ...
Unpacking python3-docker (5.0.3-1ubuntu1.1) ...
Selecting previously unselected package python3-dockerpty.
Preparing to unpack .../2-python3-dockerpty_0.4.1-5_all.deb ...
Unpacking python3-dockerpty (0.4.1-5) ...
Selecting previously unselected package python3-dockerpty.
Preparing to unpack .../3-python3-dockerpty_0.6.2-6_all.deb ...
Unpacking python3-dockerpty (0.6.2-6) ...
Selecting previously unselected package python3-dotenv.
Preparing to unpack .../4-python3-dotenv_1.0.1-1_all.deb ...
Unpacking python3-dotenv (1.0.1-1) ...
Selecting previously unselected package python3-texttable.
Preparing to unpack .../5-python3-texttable_1.6.7-1_all.deb ...
Unpacking python3-texttable (1.6.7-1) ...
Selecting previously unselected package python3-compose.
Preparing to unpack .../6-python3-compose_1.29.2-6ubuntu1_all.deb ...
Unpacking python3-compose (1.29.2-6ubuntu1) ...
Selecting previously unselected package docker-compose.
Preparing to unpack .../7-docker-compose_1.29.2-6ubuntu1_all.deb ...
Unpacking docker-compose (1.29.2-6ubuntu1) ...
Setting up python3-websocket (1.7.0-1) ...
Setting up python3-docker (5.0.3-1ubuntu1.1) ...
Setting up python3-dockerpty (0.4.1-5) ...
Setting up python3-dockerpty (0.6.2-6) ...
Setting up python3-dotenv (1.0.1-1) ...
Setting up python3-texttable (1.6.7-1) ...
Setting up python3-compose (1.29.2-6ubuntu1) ...
Setting up docker-compose (1.29.2-6ubuntu1) ...
```

Figure 6. 1 - Install docker-compose package

- The docker compose yaml file is maintained under the following directory, where all the docker images were created locally (~/.vcct/ in this instance).

A terminal window titled 'thinukam@thinukam: ~' showing the command 'sudo nano ~/.vcct/docker-compose.yml' being executed. The output shows the file being created and the directory being listed. The terminal text is as follows:

```
thinukam@thinukam:~$ sudo nano ~/.vcct/docker-compose.yml
thinukam@thinukam:~$ ls -l ~/.vcct/
total 16
drwxrwxr-x 2 thinukam docker 4096 Oct 1 19:24 dhcp-server
drwxrwxr-x 2 thinukam docker 4096 Oct 1 19:07 dns-server
-rw-r--r-- 1 root root 932 Oct 2 13:34 docker-compose.yml
drwxrwxr-x 2 thinukam docker 4096 Oct 1 20:46 proxy-server
thinukam@thinukam:~$
```

Figure 6. 2 - Directory for docker compose yaml file

- The YAML file defines all three services in a single configuration, enabling their simultaneous deployment and control.
- Each service section specifies the Docker image, container name, network mode, required capabilities, and mounted configuration or data volumes.

```

GNU nano 7.2 docker-compose.yml *
version: "2.4"

services:
  dns:
    image: it22310132/dns-server:1.0
    container_name: dns-container
    network_mode: "host"
    volumes:
      - ./dns-server/named.conf.options:/etc/bind/named.conf.options
      - ./dns-server/named.conf.local:/etc/bind/named.conf.local
      - ./dns-server/db.csne.vcct.com:/etc/bind/db.csne.vcct.com
    restart: unless-stopped

  dhcp:
    image: it22310132/dhcp-server:1.0
    container_name: dhcp-container
    network_mode: "host"
    environment:
      - INTERFACE=ens160 # replace with real interface
    cap_add:
      - NET_ADMIN
      - NET_RAW
    volumes:
      - ./dhcp-server/dhcpd.conf:/etc/dhcp/dhcpd.conf
    restart: unless-stopped

  squid:
    image: it22310132/proxy-server:1.0
    container_name: proxy-container
    network_mode: "host"
    volumes:
      - ./proxy-server/squid.conf:/etc/squid/squid.conf
      - ./proxy-server/entrypoint.sh:/entrypoint.sh
    deploy:
      resources:
        limits:
          cpus: 1
          memory: 512M
    restart: unless-stopped
  
```

Figure 6. 3 - Docker compose YAML file

- To initiate the simultaneous operation of containers, all currently active containers must be terminated and removed.

```

thinukam@thinukam:~/vcct$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
43929d81eada   it22310132/proxy-server:1.0         "/entrypoint.sh"        16 hours ago  Up 57 minutes        proxy-container
419d9d204668   it22310132/dhcp-server:1.0         "/usr/sbin/dhcpd -f ..." 16 hours ago  Up 57 minutes        dhcp-container
9dc2d5b567f3   it22310132/dns-server:1.0         "named -g -c /etc/bi..." 16 hours ago  Up 57 minutes        dns-container

thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$ docker stop dns-container dhcp-container proxy-container
dns-container
dhcp-container
proxy-container
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$ docker rm dns-container dhcp-container proxy-container
dns-container
dhcp-container
proxy-container
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
thinukam@thinukam:~/vcct$
  
```

Figure 6. 4 - Terminate and remove current active containers

- Run the YAML file with ***docker-compose*** command to simultaneously start the containers and verify the status of the containers.

```
thinukam@thinukam: ~/vcct
thinukam@thinukam:~/vcct$ docker-compose up -d
Creating proxy-container ... done
Creating dhcp-container ... done
Creating dns-container ... done
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$ docker-compose ps

```

Name	Command	State	Ports
dhcp-container	/usr/sbin/dhcpd -f -d --no ...	Up	
dns-container	named -g -c /etc/bind/name ...	Up	
proxy-container	/entrypoint.sh	Up	

```
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$
thinukam@thinukam:~/vcct$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
323cde16bd34	it22310132/dns-server:1.0	"named -g -c /etc/bi..."	14 seconds ago	Up 13 seconds		dns-container
8f43ccb52048	it22310132/dhcp-server:1.0	"/usr/sbin/dhcpd -f ..."	14 seconds ago	Up 13 seconds		dhcp-container
430023cbbe5d	it22310132/proxy-server:1.0	"/entrypoint.sh"	14 seconds ago	Up 13 seconds		proxy-container

```
thinukam@thinukam:~/vcct$
```

Figure 6. 5 - Run the containers simultaneously

Conclusion

This assignment provided practical experience in deploying essential network services, including DNS, DHCP, and Proxy, within containerized environments using Docker. Each service was successfully configured, containerized, and verified to operate correctly on the local network. The use of Docker simplified service management, improved portability, and ensured consistent configurations across deployments. Implementing Docker Compose further streamlined the process by enabling centralized control of multiple containers. Overall, the task strengthened understanding of container-based networking and demonstrated how Docker can efficiently host and manage critical infrastructure services.

References

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